



SIGNALS AND SYSTEMS

LECTURE 6

24/3/2026

CONTENT

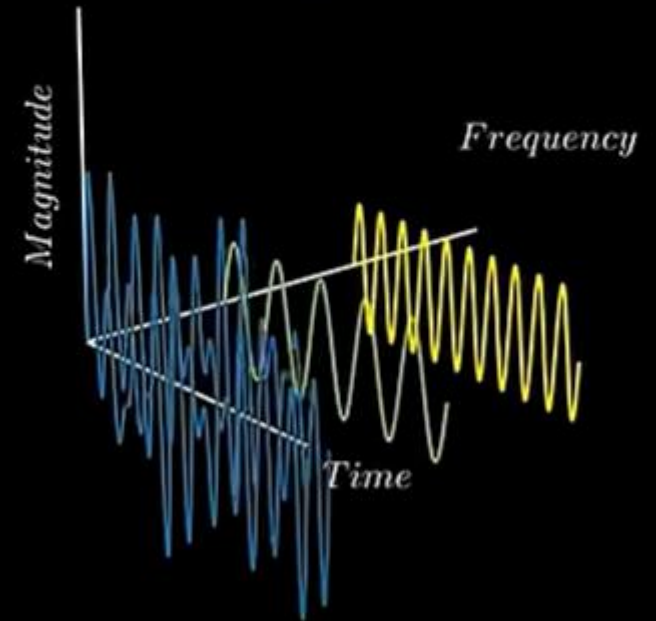
- LTI systems – Time Domain Analysis
- LTI system- Frequency Domain Analysis
- Fourier Series
- Fourier Transform
- Frequency Response of LTI Systems
- Laplace Transform



Fourier Analysis of Continuous-Time Signals and Systems

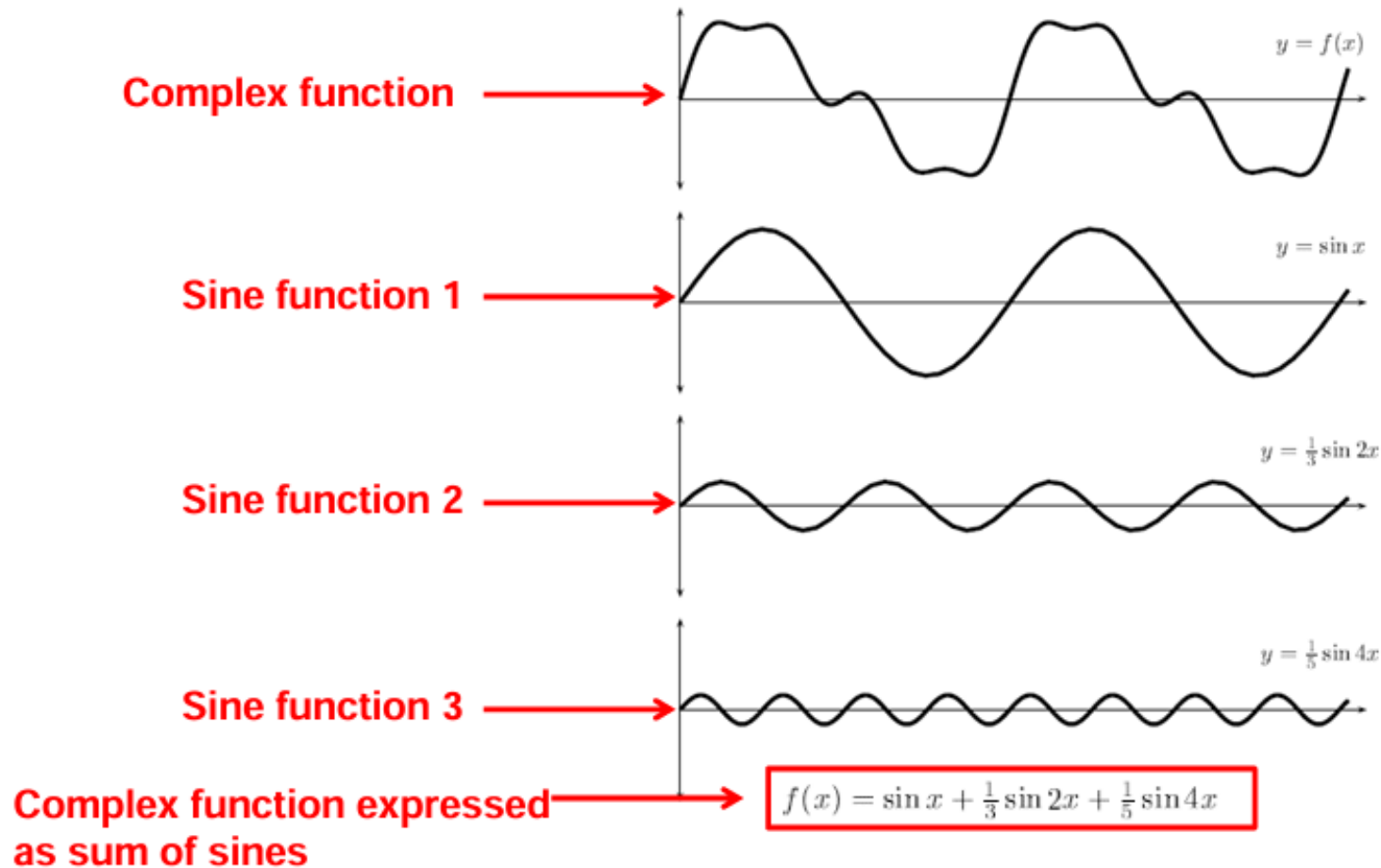
Fourier Transform

$$\mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$



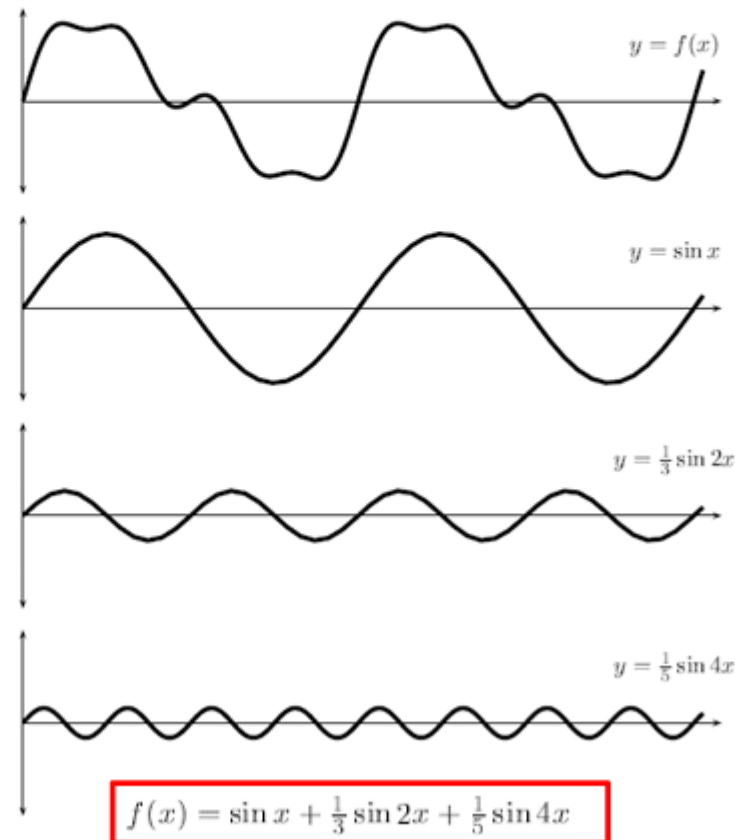
Fourier Transform

- **Main idea:** Any periodic function can be decomposed into a summation of sines and cosines



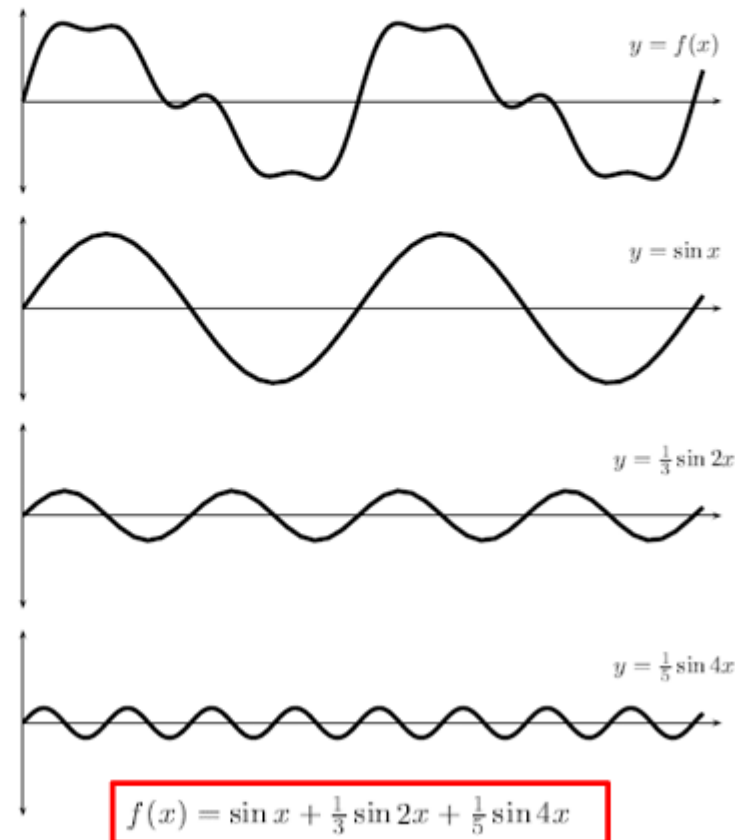
Fourier Transform: Why?

- Mathematically easier to analyze effects of transmission medium, noise, etc on simple sine functions, then add to get effect on complex signal



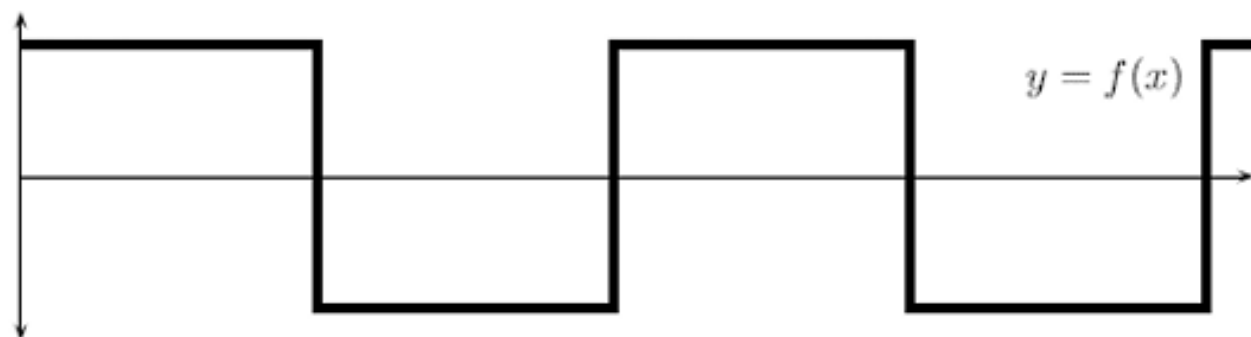
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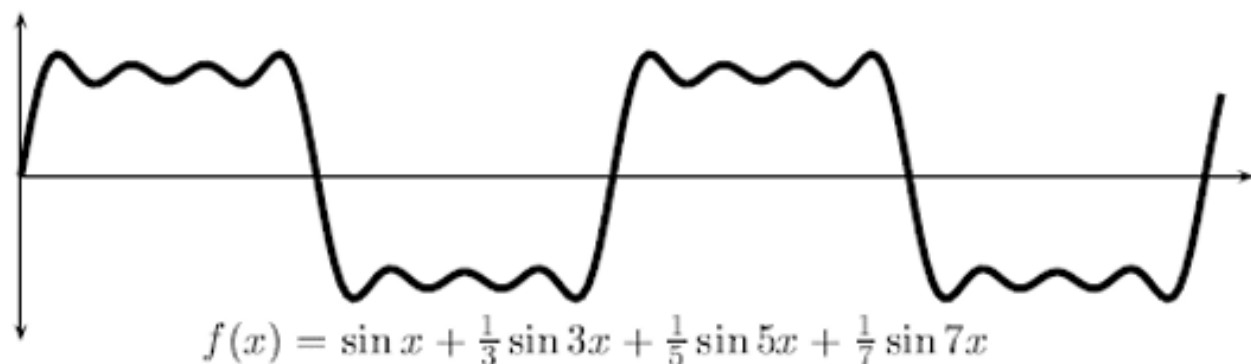


Fourier Transform: Another Example

Square wave



Approximation
Using sines

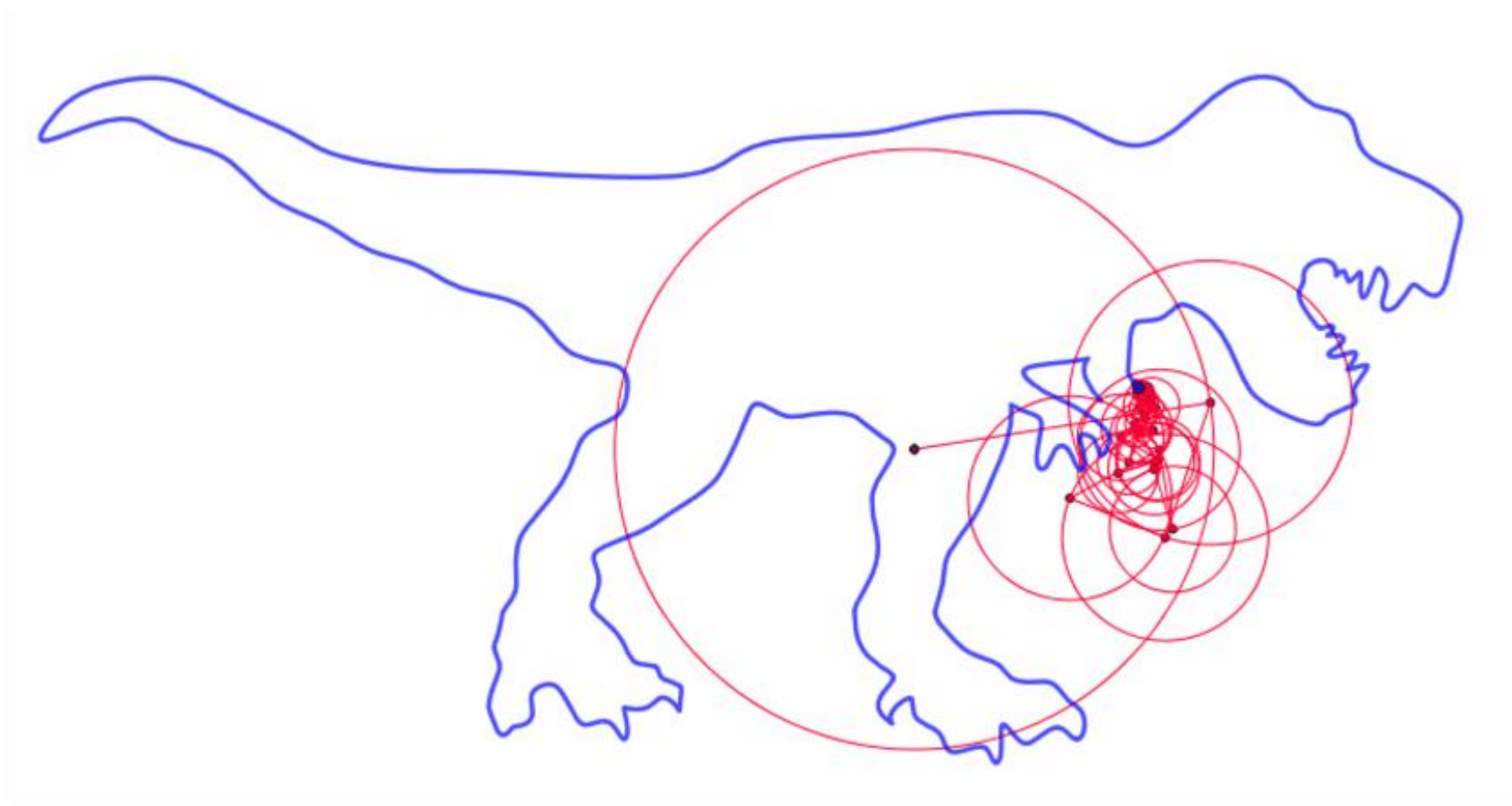


Observation 4: The sine terms go to infinity.
The more sines we add, the closer the
approximation of the original.

"Can we draw any shape using a Fourier series expansion?"



Drawing images with Fourier Series



Fourier dinosaur

- <https://medium.com/@vladimirstojoc/drawing-images-with-fourier-series-e7468ecb47b0>

$$F(\nu) = \int_{-\infty}^{\infty} \underbrace{f(t)}_{\text{Time}} e^{-2\pi i \nu t} dt$$

Fourier transform

$$f(t) = \int_{-\infty}^{\infty} \underbrace{F(\nu)}_{\text{Frequency}} e^{2\pi i \nu t} d\nu$$

Inverse Fourier transform

What is a transform? It's a mapping between domains

Time

$f(t)$

Frequency

$F(\nu)$

$\nu = \frac{\omega}{2\pi} [\text{Hz}]$

5.2 Fourier Series Representation of Periodic Signals

A. Periodic Signals:

In Chap. 1 we defined a continuous-time signal $x(t)$ to be periodic if there is a positive nonzero value of T for which

$$x(t + T) = x(t) \quad \text{all } t \quad (5.1)$$

The fundamental period T_0 of $x(t)$ is the smallest positive value of T for which Eq. (5.1) is satisfied, and $1/T_0 = f_0$ is referred to as the *fundamental frequency*.

Two basic examples of periodic signals are the real sinusoidal signal

$$x(t) = \cos(\omega_0 t + \phi) \quad (5.2)$$

and the complex exponential signal

$$x(t) = e^{j\omega_0 t} \quad (5.3)$$

where $\omega_0 = 2\pi/T_0 = 2\pi f_0$ is called the *fundamental angular frequency*.

B. Complex Exponential Fourier Series Representation:

The complex exponential Fourier series representation of a periodic signal $x(t)$ with fundamental period T_0 is given by

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \quad \omega_0 = \frac{2\pi}{T_0} \quad (5.4)$$

where c_k are known as the *complex Fourier coefficients* and are given by

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt \quad (5.5)$$

where $\int_{T_0}$ denotes the integral over any one period and 0 to T_0 or $-T_0/2$ to $T_0/2$ is commonly used for the integration. Setting $k = 0$ in Eq. (5.5), we have

$$c_0 = \frac{1}{T_0} \int_{T_0} x(t) dt \quad (5.6)$$

which indicates that c_0 equals the average value of $x(t)$ over a period.

When $x(t)$ is real, then from Eq. (5.5) it follows that

$$c_{-k} = c_k^* \quad (5.7)$$

where the asterisk indicates the complex conjugate.

C. Trigonometric Fourier Series:

The trigonometric Fourier series representation of a periodic signal $x(t)$ with fundamental period T_0 is given by

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t) \quad \omega_0 = \frac{2\pi}{T_0} \quad (5.8)$$

where a_k and b_k are the Fourier coefficients given by

$$a_k = \frac{2}{T_0} \int_{T_0} x(t) \cos k\omega_0 t \, dt \quad (5.9a)$$

$$b_k = \frac{2}{T_0} \int_{T_0} x(t) \sin k\omega_0 t \, dt \quad (5.9b)$$

The coefficients a_k and b_k and the complex Fourier coefficients c_k are related by (Prob. 5.3)

$$\frac{a_0}{2} = c_0 \quad a_k = c_k + c_{-k} \quad b_k = j(c_k - c_{-k}) \quad (5.10)$$

From Eq. (5.10) we obtain

$$c_k = \frac{1}{2}(a_k - jb_k) \quad c_{-k} = \frac{1}{2}(a_k + jb_k) \quad (5.11)$$

When $x(t)$ is real, then a_k and b_k are real and by Eq. (5.10) we have

$$a_k = 2 \operatorname{Re}[c_k] \quad b_k = -2 \operatorname{Im}[c_k] \quad (5.12)$$

Even and Odd Signals:

If a periodic signal $x(t)$ is even, then $b_k = 0$ and its Fourier series (5.8) contains only cosine terms:

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos k\omega_0 t \qquad \omega_0 = \frac{2\pi}{T_0} \qquad (5.13)$$

If $x(t)$ is odd, then $a_k = 0$ and its Fourier series contains only sine terms:

$$x(t) = \sum_{k=1}^{\infty} b_k \sin k\omega_0 t \qquad \omega_0 = \frac{2\pi}{T_0} \qquad (5.14)$$

D. Harmonic Form Fourier Series:

Another form of the Fourier series representation of a real periodic signal $x(t)$ with fundamental period T_0 is

$$x(t) = C_0 + \sum_{k=1}^{\infty} C_k \cos(k\omega_0 t - \theta_k) \quad \omega_0 = \frac{2\pi}{T_0} \quad (5.15)$$

Equation (5.15) can be derived from Eq. (5.8) and is known as the *harmonic form* Fourier series of $x(t)$. The term C_0 is known as the *dc component*, and the term $C_k \cos(k\omega_0 t - \theta_k)$ is referred to as the *kth harmonic component* of $x(t)$. The first harmonic component $C_1 \cos(\omega_0 t - \theta_1)$ is commonly called the *fundamental component* because it has the same fundamental period as $x(t)$. The coefficients C_k and the angles θ_k are called the *harmonic amplitudes* and *phase angles*, respectively, and they are related to the Fourier coefficients a_k and b_k by

$$C_0 = \frac{a_0}{2} \quad C_k = \sqrt{a_k^2 + b_k^2} \quad \theta_k = \tan^{-1} \frac{b_k}{a_k} \quad (5.16)$$

For a real periodic signal $x(t)$, the Fourier series in terms of complex exponentials as given in Eq. (5.4) is mathematically equivalent to either of the two forms in Eqs. (5.8) and (5.15). Although the latter two are common forms for Fourier series, the complex form in Eq. (5.4) is more general and usually more convenient, and we will use that form almost exclusively.

E. Convergence of Fourier Series:

It is known that a periodic signal $x(t)$ has a Fourier series representation if it satisfies the following Dirichlet conditions:

1. $x(t)$ is absolutely integrable over any period; that is,

$$\int_{T_0} |x(t)| dt < \infty \quad (5.17)$$

2. $x(t)$ has a finite number of maxima and minima within any finite interval of t .
3. $x(t)$ has a finite number of discontinuities within any finite interval of t , and each of these discontinuities is finite.

F. Amplitude and Phase Spectra of a Periodic Signal:

Let the complex Fourier coefficients c_k in Eq. (5.4) be expressed as

$$c_k = |c_k| e^{j\phi_k} \quad (5.18)$$

A plot of $|c_k|$ versus the angular frequency ω is called the *amplitude spectrum* of the periodic signal $x(t)$, and a plot of ϕ_k versus ω is called the *phase spectrum* of $x(t)$. Since the index k assumes only integers, the amplitude and phase spectra are not continuous curves but appear only at the discrete frequencies $k\omega_0$. They are therefore referred to as *discrete frequency spectra* or *line spectra*.

For a real periodic signal $x(t)$ we have $c_{-k} = c_k^*$. Thus,

$$|c_{-k}| = |c_k| \quad \phi_{-k} = -\phi_k \quad (5.19)$$

Hence, the amplitude spectrum is an even function of ω , and the phase spectrum is an odd function of ω for a real periodic signal.

G. Power Content of a Periodic Signal:

In Chap. 1 (Prob. 1.18) we introduced the average power of a periodic signal $x(t)$ over any period as

$$P = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt \quad (5.20)$$

If $x(t)$ is represented by the complex exponential Fourier series in Eq. (5.4), then it can be shown that (Prob. 5.14)

$$\frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2 \quad (5.21)$$

Equation (5.21) is called *Parseval's identity* (or *Parseval's theorem*) for the Fourier series.

5.3 The Fourier Transform

A. From Fourier Series to Fourier Transform:

Let $x(t)$ be a nonperiodic signal of finite duration; that is,

$$x(t) = 0 \quad |t| > T_1$$

Such a signal is shown in Fig. 5-1(a). Let $x_{T_0}(t)$ be a periodic signal formed by repeating $x(t)$ with fundamental period T_0 as shown in Fig. 5-1(b). If we let $T_0 \rightarrow \infty$, we have

$$\lim_{T_0 \rightarrow \infty} x_{T_0}(t) = x(t) \quad (5.22)$$

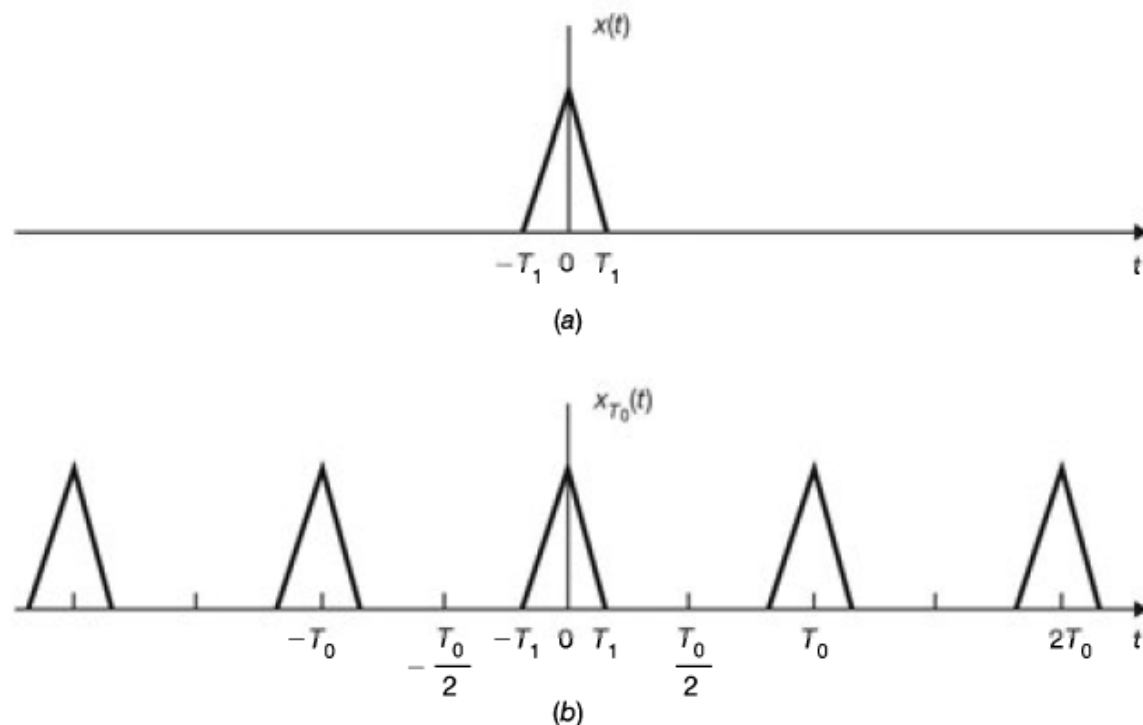


Fig. 5-1 (a) Nonperiodic signal $x(t)$; (b) periodic signal formed by periodic extension of $x(t)$.

Therefore,

$$x(t) = \lim_{T_0 \rightarrow \infty} x_{T_0}(t) = \lim_{\Delta\omega \rightarrow 0} \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\Delta\omega) e^{jk\Delta\omega t} \Delta\omega \quad (5.29)$$

The sum on the right-hand side of Eq. (5.29) can be viewed as the area under the function $X(\omega) e^{j\omega t}$, as shown in Fig. 5-2. Therefore, we obtain

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (5.30)$$

which is the Fourier representation of a nonperiodic $x(t)$.

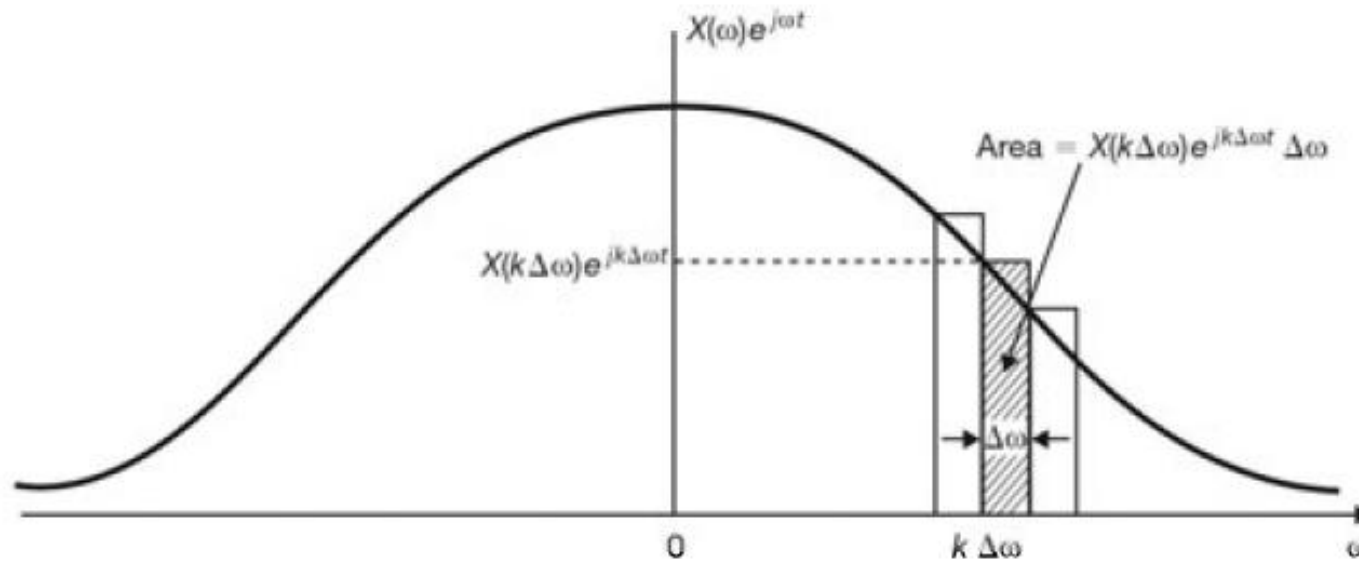


Fig. 5-2 Graphical interpretation of Eq. (5.29).

B. Fourier Transform Pair:

The function $X(\omega)$ defined by Eq. (5.25) is called the *Fourier transform* of $x(t)$, and Eq. (5.30) defines the *inverse Fourier transform of $X(\omega)$* . Symbolically they are denoted by

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{j\omega t} dt \quad (5.31)$$

$$x(t) = \mathcal{F}^{-1}\{X(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (5.32)$$

and we say that $x(t)$ and $X(\omega)$ form a Fourier transform pair denoted by

$$x(t) \leftrightarrow X(\omega) \quad (5.33)$$

C. Fourier Spectra:

The Fourier transform $X(\omega)$ of $x(t)$ is, in general, complex, and it can be expressed as

$$X(\omega) = |X(\omega)| e^{j\phi(\omega)} \quad (5.34)$$

By analogy with the terminology used for the complex Fourier coefficients of a periodic signal $x(t)$, the Fourier transform $X(\omega)$ of a nonperiodic signal $x(t)$ is the frequency-domain specification of $x(t)$ and is referred to as the *spectrum* (or *Fourier spectrum*) of $x(t)$. The quantity $|X(\omega)|$ is called the *magnitude spectrum* of $x(t)$, and $\phi(\omega)$ is called the *phase spectrum* of $x(t)$.

If $x(t)$ is a real signal, then from Eq. (5.31) we get

$$X(-\omega) = \int_{-\infty}^{\infty} x(t) e^{j\omega t} dt \quad (5.35)$$

Then it follows that

$$X(-\omega) = X^*(\omega) \quad (5.36a)$$

$$\text{and} \quad |X(-\omega)| = |X(\omega)| \quad \phi(-\omega) = -\phi(\omega) \quad (5.36b)$$

Hence, as in the case of periodic signals, the amplitude spectrum $|X(\omega)|$ is an even function and the phase spectrum $\phi(\omega)$ is an odd function of ω .

D. Convergence of Fourier Transforms:

Just as in the case of periodic signals, the sufficient conditions for the convergence of $X(\omega)$ are the following (again referred to as the Dirichlet conditions):

1. $x(t)$ is absolutely integrable; that is,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty \quad (5.37)$$

2. $x(t)$ has a finite number of maxima and minima within any finite interval.
3. $x(t)$ has a finite number of discontinuities within any finite interval, and each of these discontinuities is finite.

5.4 Properties of the Continuous-Time Fourier Transform

Basic properties of the Fourier transform are presented in the following. Many of these properties are similar to those of the Laplace transform (see Sec. 3.4).

A. Linearity:

$$a_1 x_1(t) + a_2 x_2(t) \leftrightarrow a_1 X_1(\omega) + a_2 X_2(\omega) \quad (5.49)$$

B. Time Shifting:

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(\omega) \quad (5.50)$$

Equation (5.50) shows that the effect of a shift in the time domain is simply to add a linear term $-\omega t_0$ to the original phase spectrum $\theta(\omega)$. This is known as a *linear phase shift* of the Fourier transform $X(\omega)$.

C. Frequency Shifting:

$$e^{j\omega_0 t} x(t) \leftrightarrow X(\omega - \omega_0) \quad (5.51)$$

The multiplication of $x(t)$ by a complex exponential signal $e^{j\omega_0 t}$ is sometimes called *complex modulation*. Thus, Eq. (5.51) shows that complex modulation in the time domain corresponds to a shift of $X(\omega)$ in the frequency domain. Note that the frequency-shifting property Eq. (5.51) is the dual of the time-shifting property Eq. (5.50).

D. Time Scaling:

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right) \quad (5.52)$$

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$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right) \quad (5.52)$$

where a is a real constant. This property follows directly from the definition of the Fourier transform. Equation (5.52) indicates that scaling the time variable t by the factor a causes an inverse scaling of the frequency variable ω by $1/a$, as well as an amplitude scaling of $X(\omega/a)$ by $1/|a|$. Thus, the scaling property (5.52) implies that time compression of a signal ($a > 1$) results in its spectral expansion and that time expansion of the signal ($a < 1$) results in its spectral compression.

E. Time Reversal:

$$x(-t) \leftrightarrow X(-\omega) \quad (5.53)$$

Thus, time reversal of $x(t)$ produces a like reversal of the frequency axis for $X(\omega)$. Equation (5.53) is readily obtained by setting $a = -1$ in Eq. (5.52).

F. Duality (or Symmetry):

$$X(t) \leftrightarrow 2\pi x(-\omega) \quad (5.54)$$

The duality property of the Fourier transform has significant implications. This property allows us to obtain both of these dual Fourier transform pairs from one evaluation of Eq. (5.31) (Probs. 5.20 and 5.22).

G. Differentiation in the Time Domain:

$$\frac{dx(t)}{dt} \leftrightarrow j\omega X(\omega) \quad (5.55)$$

Equation (5.55) shows that the effect of differentiation in the time domain is the multiplication of $X(\omega)$ by $j\omega$ in the frequency domain (Prob. 5.28).

H. Differentiation in the Frequency Domain:

$$(-jt)x(t) \leftrightarrow \frac{dX(\omega)}{d\omega} \quad (5.56)$$

Equation (5.56) is the dual property of Eq. (5.55).

I. Integration in the Time Domain:

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \pi X(0) \delta(\omega) + \frac{1}{j\omega} X(\omega) \quad (5.57)$$

Since integration is the inverse of differentiation, Eq. (5.57) shows that the frequency domain operation corresponding to time-domain integration is multiplication by $1/j\omega$, but an additional term is needed to account for a possible dc component in the integrator output. Hence, unless $X(0) = 0$, a dc component is produced by the integrator (Prob. 5.33).

J. Convolution:

$$x_1(t) * x_2(t) \leftrightarrow X_1(\omega) X_2(\omega) \quad (5.58)$$

Equation (5.58) is referred to as the *time convolution theorem*, and it states that convolution in the time domain becomes multiplication in the frequency domain (Prob. 5.31). As in the case of the Laplace transform, this convolution property plays an important role in the study of continuous-time LTI systems (Sec. 5.5) and also forms the basis for our discussion of filtering (Sec. 5.6).

K. Multiplication:

$$x_1(t)x_2(t) \leftrightarrow \frac{1}{2\pi} X_1(\omega) * X_2(\omega) \quad (5.59)$$

The multiplication property (5.59) is the dual property of Eq. (5.58) and is often referred to as the *frequency convolution theorem*. Thus, multiplication in the time domain becomes convolution in the frequency domain (Prob. 5.35).

L. Additional Properties:

If $x(t)$ is real, let

$$x(t) = x_e(t) + x_o(t) \quad (5.60)$$

where $x_e(t)$ and $x_o(t)$ are the even and odd components of $x(t)$, respectively. Let

$$x(t) \leftrightarrow X(\omega) = A(\omega) + jB(\omega)$$

Then

$$X(-\omega) = X^*(\omega) \quad (5.61a)$$

$$x_e(t) \leftrightarrow \operatorname{Re}\{X(\omega)\} = A(\omega) \quad (5.61b)$$

$$x_o(t) \leftrightarrow j \operatorname{Im}\{X(\omega)\} = jB(\omega) \quad (5.61c)$$

TABLE 5-1 Properties of the Fourier Transform

PROPERTY	SIGNAL	FOURIER TRANSFORM
	$x(t)$	$X(\omega)$
	$x_1(t)$	$X_1(\omega)$
	$x_2(t)$	$X_2(\omega)$
Linearity	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(\omega) + a_2X_2(\omega)$
Time shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(\omega)$
Frequency shifting	$e^{j\omega_0 t} x(t)$	$X(\omega - \omega_0)$
Time scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{\omega}{a}\right)$
Time reversal	$x(-t)$	$X(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Time differentiation	$\frac{dx(t)}{dt}$	$j\omega X(\omega)$
Frequency differentiation	$(-jt)x(t)$	$\frac{dX(\omega)}{d\omega}$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\pi X(0) \delta(\omega) + \frac{1}{j\omega} X(\omega)$
Convolution	$x_1(t) * x_2(t)$	$X_1(\omega)X_2(\omega)$
Multiplication	$x_1(t)x_2(t)$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
Real signal	$x(t) = x_e(t) + x_o(t)$	$X(\omega) = A(\omega) + jB(\omega)$ $X(-\omega) = X^*(\omega)$
Even component	$x_e(t)$	$\text{Re}\{X(\omega)\} = A(\omega)$
Odd component	$x_o(t)$	$j \text{Im}\{X(\omega)\} = jB(\omega)$
Parseval's relations	$\int_{-\infty}^{\infty} x_1(\lambda)X_2(\lambda) d\lambda = \int_{-\infty}^{\infty} X_1(\lambda)x_2(\lambda) d\lambda$	

TABLE 5.2 Common Fourier Transforms Pairs

$x(t)$	$X(\omega)$
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$-j\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$u(-t)$	$\pi\delta(\omega) - \frac{1}{j\omega}$
$e^{-at}u(t), a > 0$	$\frac{1}{j\omega + a}$
$te^{-at}u(t), a > 0$	$\frac{1}{(j\omega + a)^2}$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$
$\frac{1}{a^2 + t^2}$	$e^{-a \omega }$
$e^{-at^2}, a > 0$	$\sqrt{\frac{\pi}{a}} e^{-\omega^2/4a}$

5.5 The Frequency Response of Continuous-Time LTI Systems

A. Frequency Response:

In Sec. 2.2 we showed that the output $y(t)$ of a continuous-time LTI system equals the convolution of the input $x(t)$ with the impulse response $h(t)$; that is,

$$y(t) = x(t) * h(t) \quad (5.65)$$

Applying the convolution property (5.58), we obtain

$$Y(\omega) = X(\omega)H(\omega) \quad (5.66)$$

where $Y(\omega)$, $X(\omega)$, and $H(\omega)$ are the Fourier transforms of $y(t)$, $x(t)$, and $h(t)$, respectively. From Eq. (5.66) we have

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \quad (5.67)$$

The function $H(\omega)$ is called the *frequency response* of the system. Relationships represented by Eqs. (5.65) and (5.66) are depicted in Fig. 5-3. Let

$$H(\omega) = |H(\omega)| e^{j\theta_H(\omega)} \quad (5.68)$$

Then $|H(\omega)|$ is called the *magnitude response* of the system, and $\theta_H(\omega)$ the *phase response* of the system.

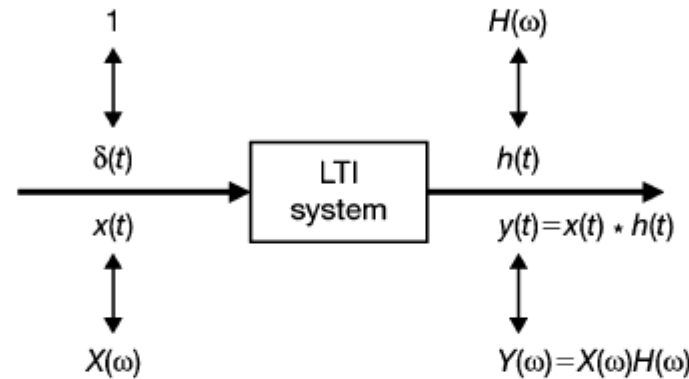


Fig. 5-3 Relationships between inputs and outputs in an LTI system.

Consider the complex exponential signal

$$x(t) = e^{j\omega_0 t} \quad (5.69)$$

with Fourier transform (Prob. 5.23)

$$X(\omega) = 2\pi\delta(\omega - \omega_0) \quad (5.70)$$

Then from Eqs. (5.66) and (1.26) we have

$$Y(\omega) = 2\pi H(\omega_0) \delta(\omega - \omega_0) \quad (5.71)$$

Taking the inverse Fourier transform of $Y(\omega)$, we obtain

$$y(t) = H(\omega_0) e^{j\omega_0 t} \quad (5.72)$$

which indicates that the complex exponential signal $e^{j\omega_0 t}$ is an eigenfunction of the LTI system with corresponding eigenvalue $H(\omega_0)$, as previously observed in Chap. 2 (Sec. 2.4 and Prob. 2.17]. Furthermore, by the linearity property (5.49), if the input $x(t)$ is periodic with the Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \quad (5.73)$$

then the corresponding output $y(t)$ is also periodic with the Fourier series

$$y(t) = \sum_{k=-\infty}^{\infty} c_k H(k\omega_0) e^{jk\omega_0 t} \quad (5.74)$$

If $x(t)$ is not periodic, then from Eq. (5.30)

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (5.75)$$

and using Eq. (5.66), the corresponding output $y(t)$ can be expressed as

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega) X(\omega) e^{j\omega t} d\omega \quad (5.76)$$

Thus, the behavior of a continuous-time LTI system in the frequency domain is completely characterized by its frequency response $H(\omega)$. Let

$$X(\omega) = |X(\omega)| e^{j\theta_X(\omega)} \quad Y(\omega) = |Y(\omega)| e^{j\theta_Y(\omega)} \quad (5.77)$$

5.4. Determine the complex exponential Fourier series representation for each of the following signals:

(a) $x(t) = \cos \omega_0 t$

(b) $x(t) = \sin \omega_0 t$

(c) $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$

(d) $x(t) = \cos 4t + \sin 6t$

(e) $x(t) = \sin^2 t$

(a) Rather than using Eq. (5.5) to evaluate the complex Fourier coefficients c_k using Euler's formula, we get

$$\cos \omega_0 t = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) = \frac{1}{2} e^{-j\omega_0 t} + \frac{1}{2} e^{j\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Thus, the complex Fourier coefficients for $\cos \omega_0 t$ are

$$c_1 = \frac{1}{2} \qquad c_{-1} = \frac{1}{2} \qquad c_k = 0, |k| \neq 1$$

(b) In a similar fashion we have

$$\sin \omega_0 t = \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t}) = -\frac{1}{2j} e^{-j\omega_0 t} + \frac{1}{2j} e^{j\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Thus, the complex Fourier coefficients for $\sin \omega_0 t$ are

$$c_1 = \frac{1}{2j} \quad c_{-1} = -\frac{1}{2j} \quad c_k = 0, |k| \neq 1$$

(c) The fundamental angular frequency ω_0 of $x(t)$ is 2. Thus,

$$x(t) = \cos\left(2t + \frac{\pi}{4}\right) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt}$$

Now

$$\begin{aligned} x(t) &= \cos\left(2t + \frac{\pi}{4}\right) = \frac{1}{2}(e^{j(2t+\pi/4)} + e^{-j(2t+\pi/4)}) \\ &= \frac{1}{2}e^{-j\pi/4}e^{-j2t} + \frac{1}{2}e^{j\pi/4}e^{j2t} = \sum_{k=-\infty}^{\infty} c_k e^{j2kt} \end{aligned}$$

Thus, the complex Fourier coefficients for $\cos(2t + \pi/4)$ are

$$\begin{aligned} c_1 &= \frac{1}{2}e^{j\pi/4} = \frac{1}{2} \frac{1+j}{\sqrt{2}} = \frac{\sqrt{2}}{4}(1+j) \\ c_{-1} &= \frac{1}{2}e^{-j\pi/4} = \frac{1}{2} \frac{1-j}{\sqrt{2}} = \frac{\sqrt{2}}{4}(1-j) \\ c_k &= 0 \quad |k| \neq 1 \end{aligned}$$

5.21. Find the Fourier transform of the signal [Fig. 5-18(a)]

$$x(t) = e^{-a|t|} \quad a > 0$$

Signal $x(t)$ can be rewritten as

$$x(t) = e^{-a|t|} = \begin{cases} e^{-at} & t > 0 \\ e^{at} & t < 0 \end{cases}$$

Then

$$\begin{aligned} X(\omega) &= \int_{-\infty}^0 e^{at} e^{-j\omega t} dt + \int_0^{\infty} e^{-at} e^{-j\omega t} dt \\ &= \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt \\ &= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} = \frac{2a}{a^2 + \omega^2} \end{aligned}$$

Hence, we get

$$e^{-a|t|} \leftrightarrow \frac{2a}{a^2 + \omega^2} \quad ($$

The Fourier transform $X(\omega)$ of $x(t)$ is shown in Fig. 5-18(b).

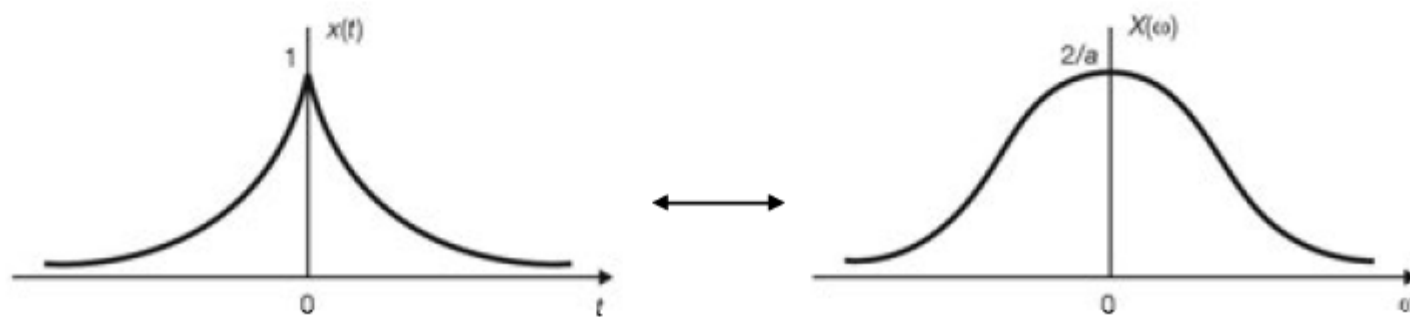


Fig. 5-18 $e^{-a|t|}$ and its Fourier transform.

5.23. Find the Fourier transforms of the following signals:

(a) $x(t) = 1$

(b) $x(t) = e^{j\omega_0 t}$

(c) $x(t) = e^{-j\omega_0 t}$

(d) $x(t) = \cos \omega_0 t$

(e) $x(t) = \sin \omega_0 t$

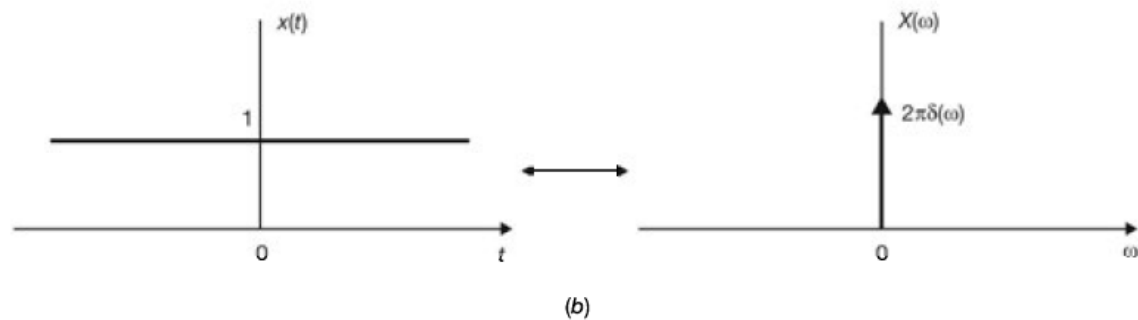
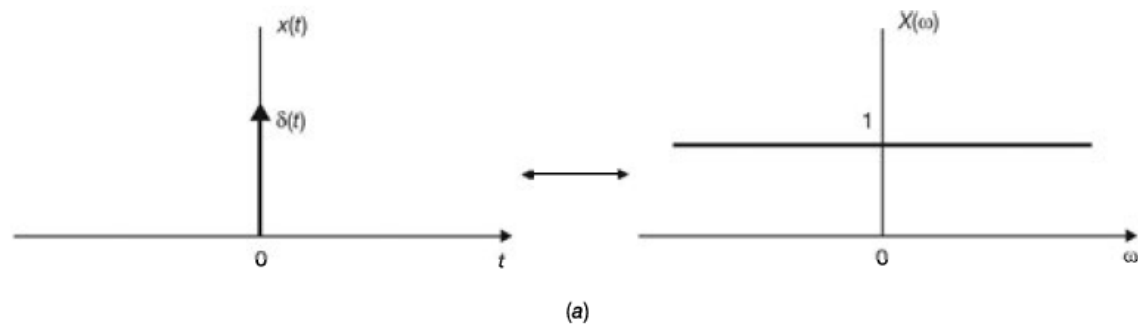
(a) By Eq. (5.43) we have

$$\delta(t) \leftrightarrow 1 \quad (5.140)$$

Thus, by the duality property (5.54) we get

$$1 \leftrightarrow 2\pi\delta(-\omega) = 2\pi\delta(\omega) \quad (5.141)$$

Figs. 5-20(a) and (b) illustrate the relationships in Eqs. (5.140) and (5.141), respectively.



Laplace Transform

E. Connection between the Fourier Transform and the Laplace Transform:

Equation (5.31) defines the Fourier transform of $x(t)$ as

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (5.38)$$

Laplace Transform

The bilateral Laplace transform of $x(t)$, as defined in Eq. (4.3), is given by

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad (5.39)$$

Comparing Eqs. (5.38) and (5.39), we see that the Fourier transform is a special case of the Laplace transform in which $s = j\omega$; that is,

$$X(s)|_{s=j\omega} = \mathcal{F}\{x(t)\} \quad (5.40)$$

Setting $s = \sigma + j\omega$ in Eq. (5.39), we have

$$X(\sigma + j\omega) = \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt = \int_{-\infty}^{\infty} [x(t) e^{-\sigma t}] e^{-j\omega t} dt$$

or

$$X(\sigma + j\omega) = \mathcal{F}\{x(t) e^{-\sigma t}\} \quad (5.41)$$

which indicates that the bilateral Laplace transform of $x(t)$ can be interpreted as the Fourier transform of $x(t) e^{-\sigma t}$.

Laplace Transform

Since the Laplace transform may be considered a generalization of the Fourier transform in which the frequency is generalized from $j\omega$ to $s = \sigma + j\omega$, the complex variable s is often referred to as the *complex frequency*.

Note that since the integral in Eq. (5.39) is denoted by $X(s)$, the integral in Eq. (5.38) may be denoted as $X(j\omega)$. Thus, in the remainder of this book both $X(\omega)$ and $X(j\omega)$ mean the same thing whenever we connect the Fourier transform with the Laplace transform. Because the Fourier transform is the Laplace transform with $s = j\omega$, it should not be assumed automatically that the Fourier transform of a signal $x(t)$ is the Laplace transform with s replaced by $j\omega$. If $x(t)$ is absolutely integrable, that is, if $x(t)$ satisfies condition (5.37), the Fourier transform of $x(t)$ can be obtained from the Laplace transform of $x(t)$ with $s = j\omega$. This is not generally true of signals which are not absolutely integrable. The following examples illustrate the above statements.

EXAMPLE 5.1 Consider the unit impulse function $\delta(t)$.

From Eq. (3.13) the Laplace transform of $\delta(t)$ is

$$\mathcal{L}\{\delta(t)\} = 1 \quad \text{all } s \quad (5.42)$$

By definitions (5.31) and (1.20) the Fourier transform of $\delta(t)$ is

$$\mathcal{F}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = 1 \quad (5.43)$$

Thus, the Laplace transform and the Fourier transform of $\delta(t)$ are the same.

EXAMPLE 5.2 Consider the exponential signal

$$x(t) = e^{-at}u(t) \quad a > 0$$

From Eq. (3.8) the Laplace transform of $x(t)$ is given by

$$\mathcal{L}\{x(t)\} = X(s) = \frac{1}{s+a} \quad \text{Re}(s) > -a \quad (5.44)$$

By definition (5.31) the Fourier transform of $x(t)$ is

$$\begin{aligned} \mathcal{F}\{x(t)\} &= X(\omega) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt \\ &= \int_{0+}^{\infty} e^{-(a+j\omega)t} dt = \frac{1}{a+j\omega} \end{aligned} \quad (5.45)$$

Thus, comparing Eqs. (5.44) and (5.45), we have

$$X(\omega) = X(s)|_{s=j\omega} \quad (5.46)$$

Note that $x(t)$ is absolutely integrable.

EXAMPLE 5.3 Consider the unit step function $u(t)$.

From Eq. (3.14) the Laplace transform of $u(t)$ is

$$\mathcal{L}\{u(t)\} = \frac{1}{s} \quad \text{Re}(s) > 0 \quad (5.47)$$

The Fourier transform of $u(t)$ is given by (Prob. 5.30)

$$\mathcal{F}\{u(t)\} = \pi\delta(\omega) + \frac{1}{j\omega} \quad (5.48)$$

Thus, the Fourier transform of $u(t)$ cannot be obtained from its Laplace transform. Note that the unit step function $u(t)$ is not absolutely integrable.

Fourier Transform (FT)

For a non-periodic signal $x(t)$:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

The inverse transform:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$$

Interpretation:

- $X(\omega)$ tells the **amount of frequency** ω in the signal.

Unlike Fourier Series:

- spectrum is **continuous**

Example 2 — Fourier Transform of an Exponential Signal

Let

$$x(t) = e^{-at}u(t), \quad a > 0$$

Compute the Fourier Transform:

$$X(\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

Combine exponents:

$$X(\omega) = \int_0^{\infty} e^{-(a+j\omega)t} dt$$

Integrating:

$$X(\omega) = \frac{1}{a + j\omega}$$

Result:

$$X(\omega) = \frac{1}{a + j\omega}$$

Example 3 — Fourier Transform of a Rectangular Pulse

Consider

$$x(t) = \begin{cases} 1, & |t| < T/2 \\ 0, & \text{otherwise} \end{cases}$$

Fourier Transform:

$$X(\omega) = \int_{-T/2}^{T/2} e^{-j\omega t} dt$$

Result:

$$X(\omega) = T \frac{\sin(\omega T/2)}{\omega T/2}$$

This is the **sinc function**.

10. Fourier Series vs Fourier Transform

Feature	Fourier Series	Fourier Transform
Signal type	Periodic	Non-periodic
Spectrum	Discrete	Continuous
Frequencies	Harmonics $k\omega_0$	All frequencies
Representation	Sum	Integral
Output	Coefficients c_k	Function $X(\omega)$

Why These Concepts Matter for Systems

For LTI systems:

Input:

$$x(t) = e^{j\omega t}$$

Output:

$$y(t) = H(\omega)e^{j\omega t}$$

Thus the system modifies each frequency by:

- amplitude scaling
- phase shift

This function

$$H(\omega)$$

is called the **frequency response** of the system.

So the full procedure becomes:

1. Decompose signal into frequencies (Fourier Transform)
2. Multiply by system response
3. Transform back to time domain

Definition

-Fourier Basis $\rightarrow \boxed{e^{j\omega t}}$

-The Fourier Transform of $x(t)$ is the projection of this function onto the Fourier basis.

-The Fourier Transform of $x(t)$ is defined as:

$$F(x(t)) = \boxed{X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt}$$

$$x(t) \overset{F}{\leftrightarrow} X(\omega)$$

-It follows that the Inverse Fourier Transform of $X(\omega)$ is:

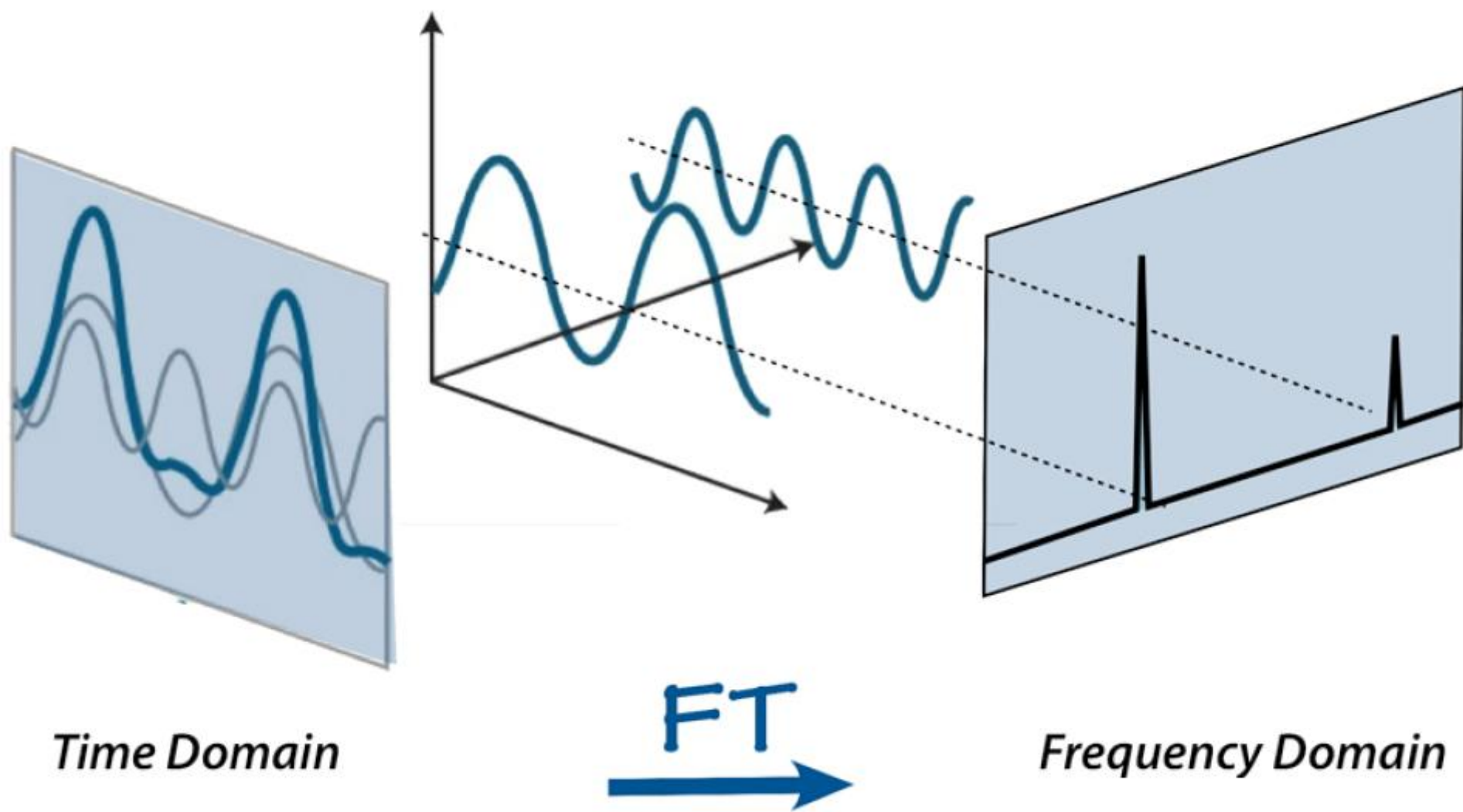
$$\mathcal{F}^{-1}(X(\omega)) = \boxed{x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega}$$

$$X(\omega) \overset{\mathcal{F}^{-1}}{\longleftrightarrow} x(t)$$

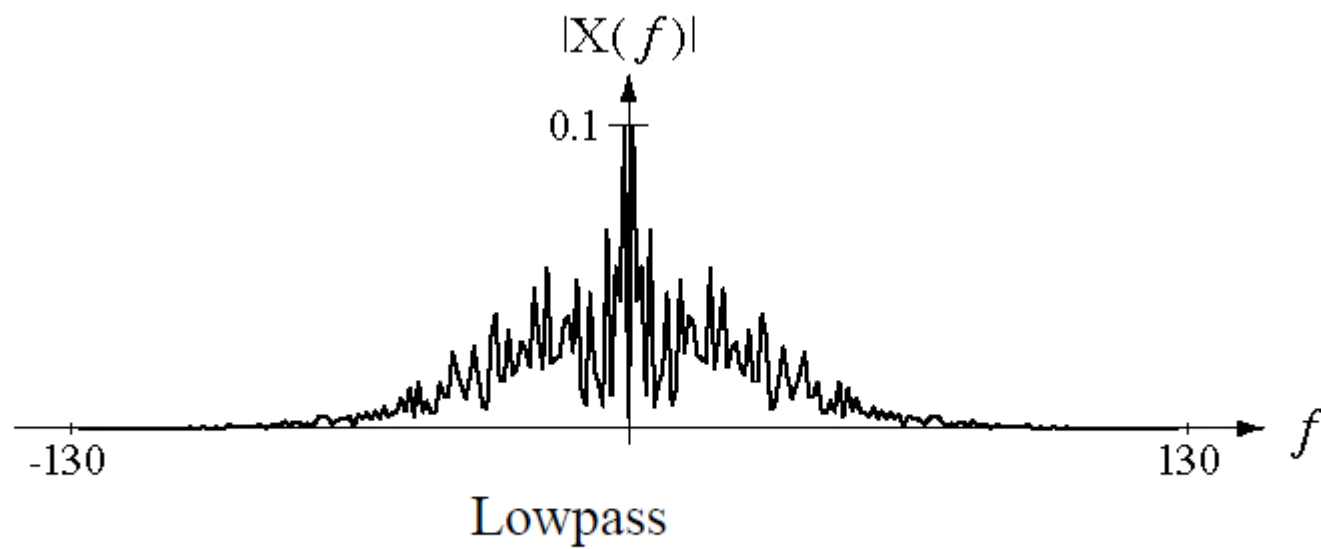
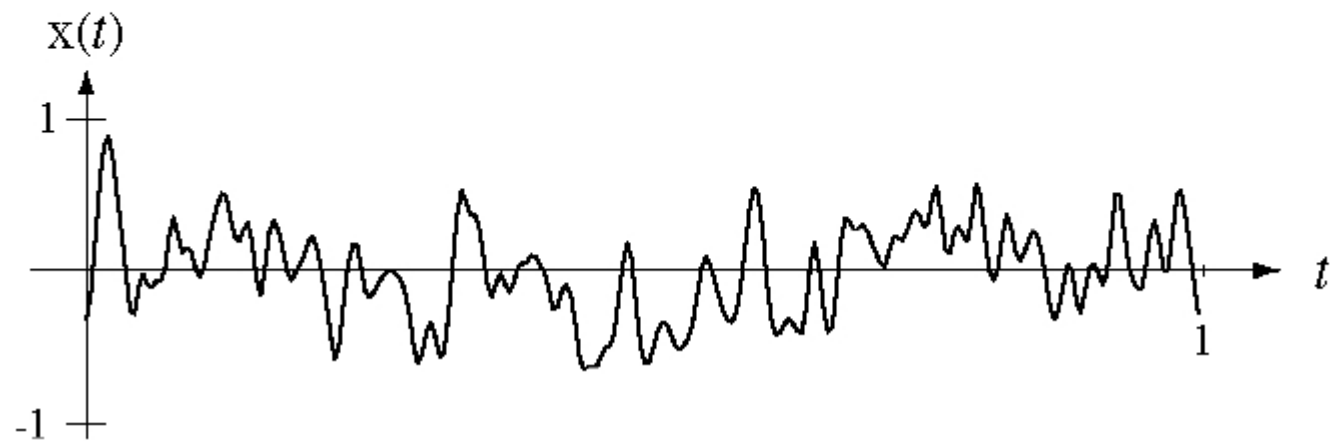
-The definitions based on frequency terms are:

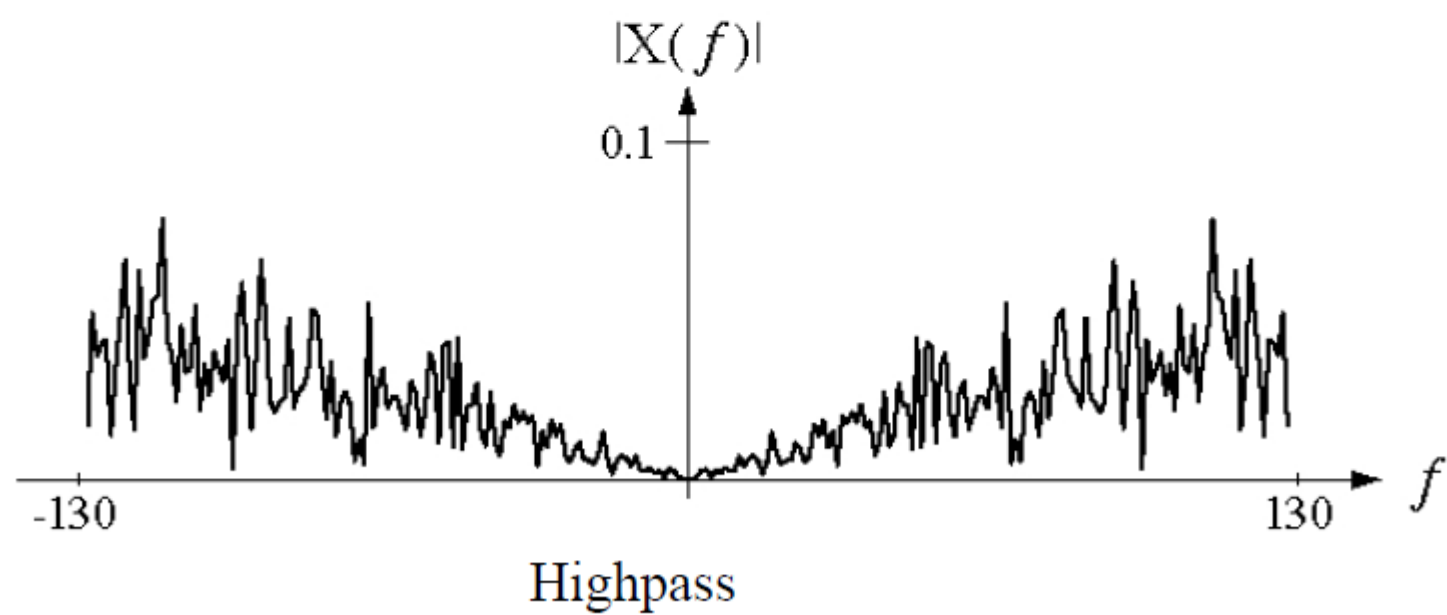
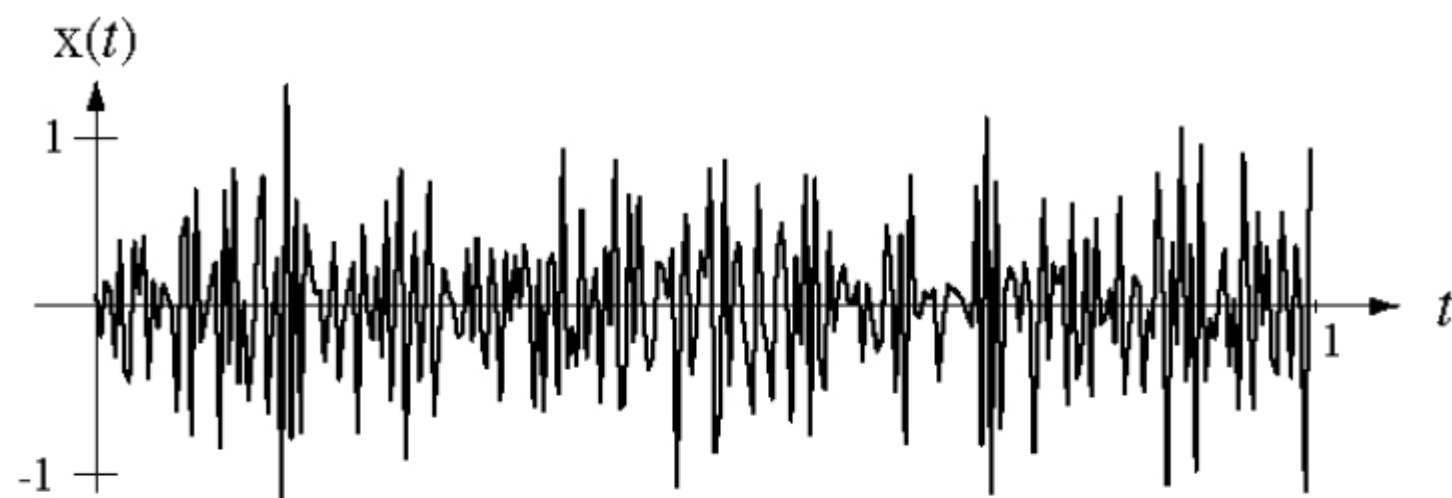
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

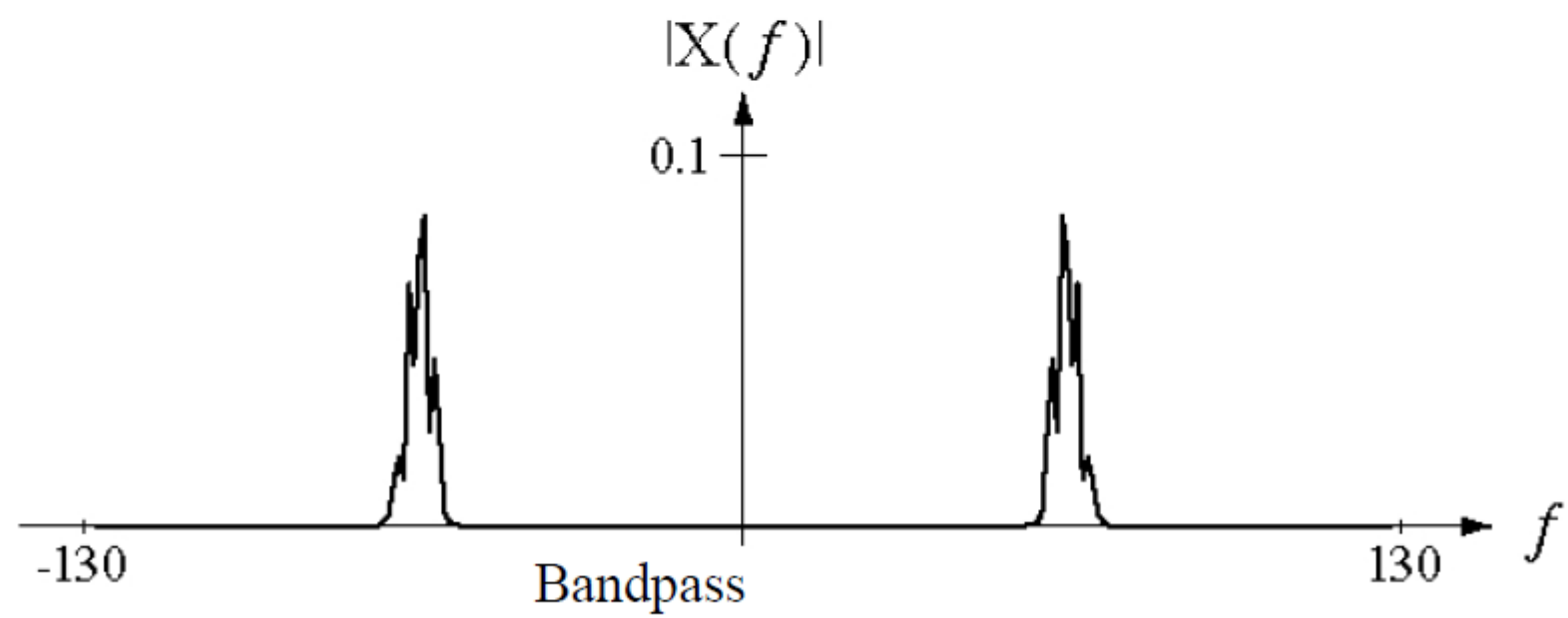
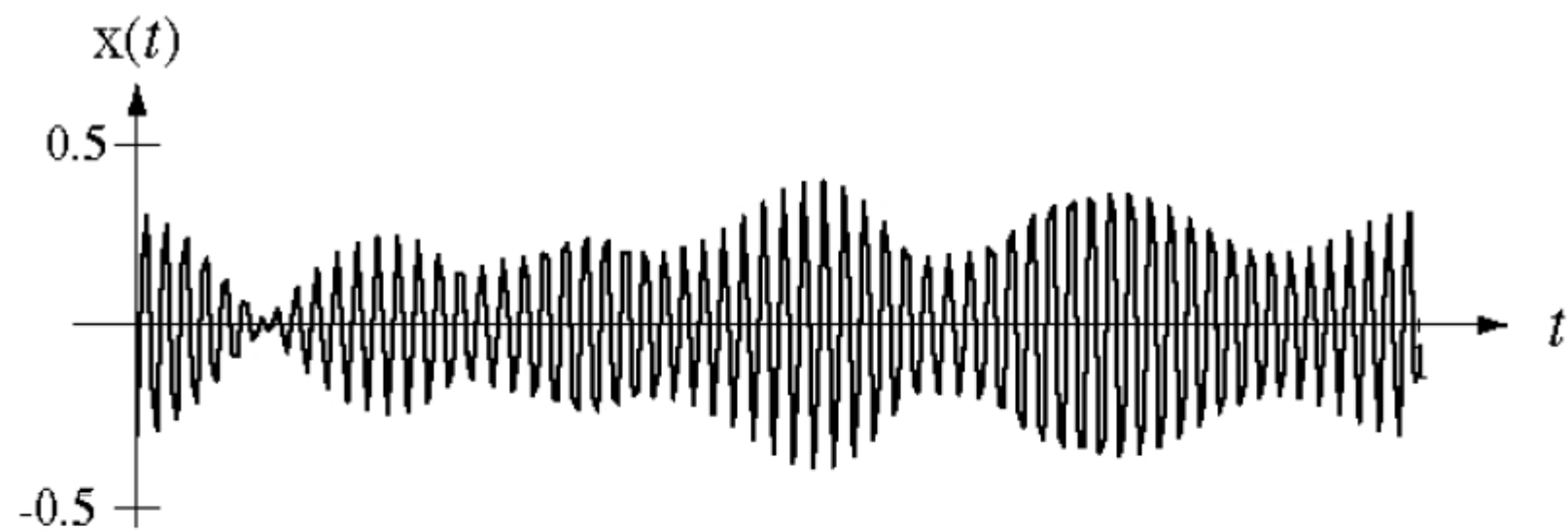
$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j\omega t} df$$



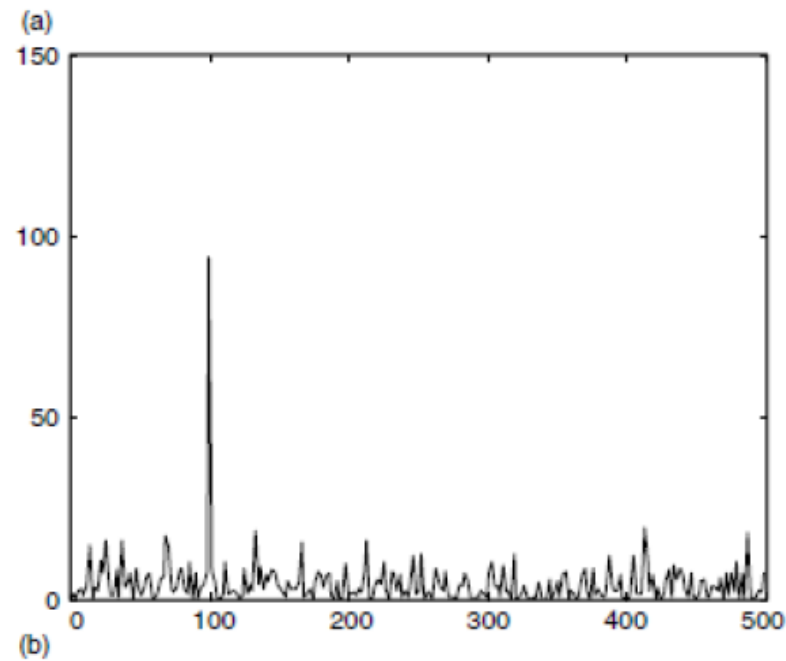
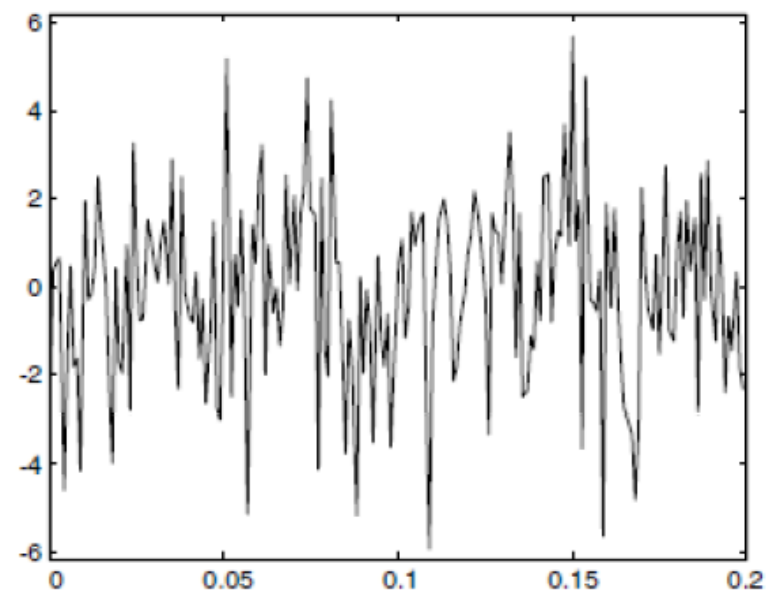
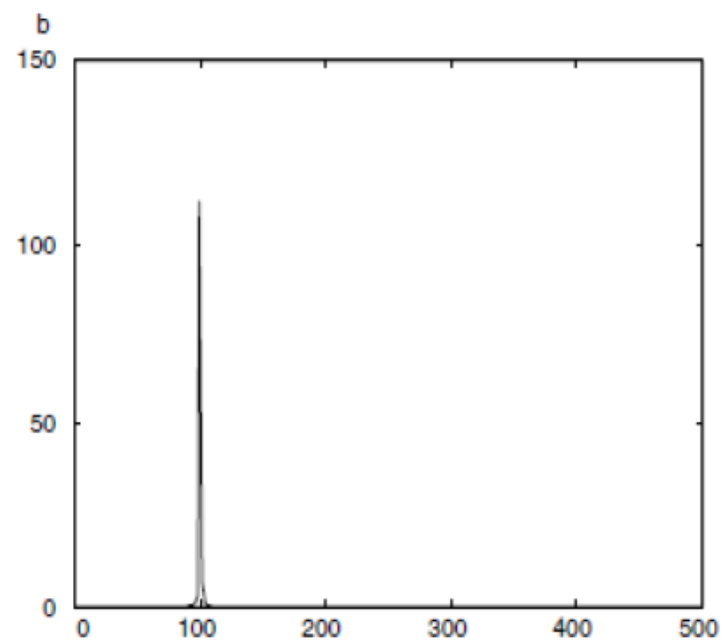
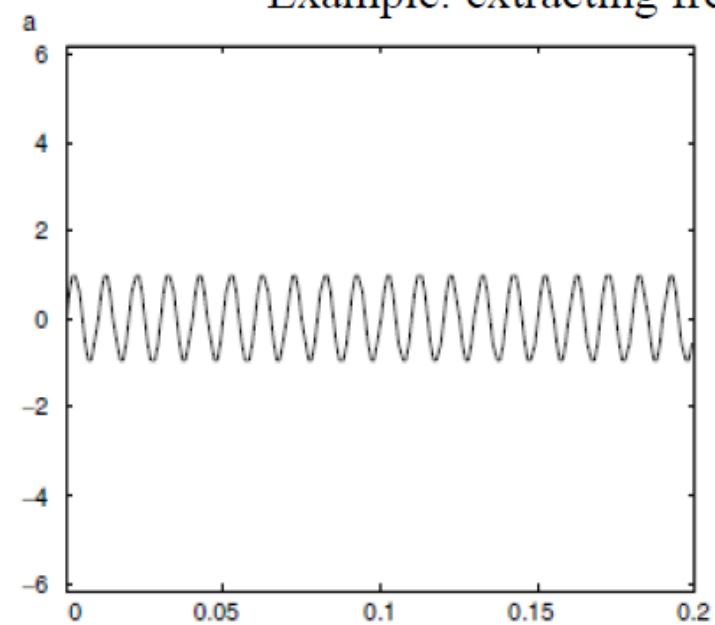
Examples



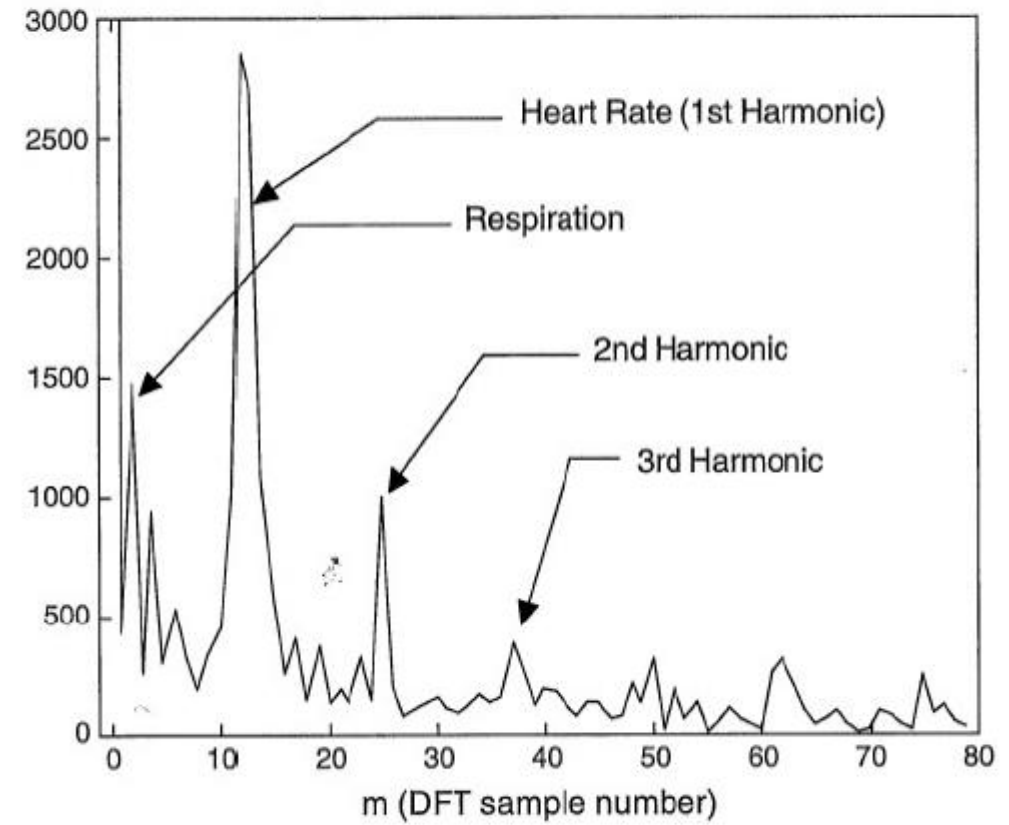
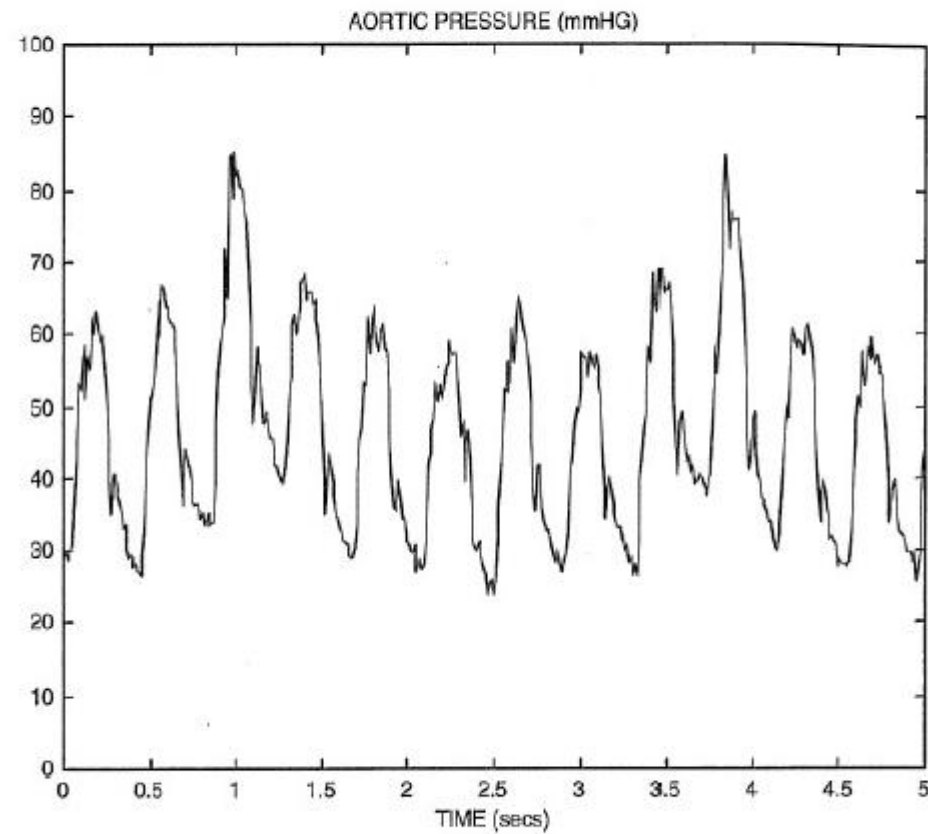




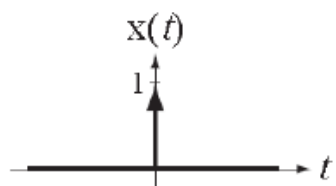
Example: extracting frequency-domain information in the signal



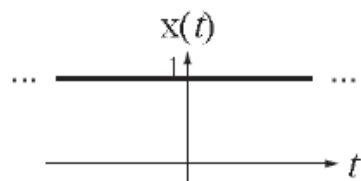
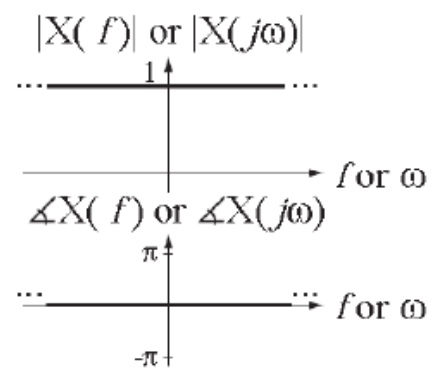
Example: blood pressure waveform (sampled at 200 points/s)



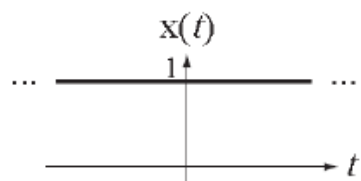
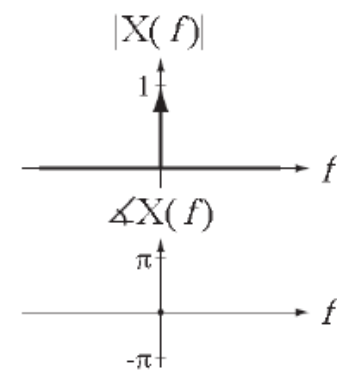
FT Pairs



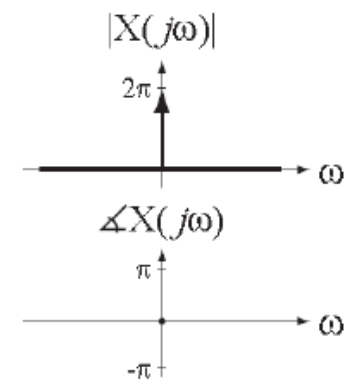
$$\delta(t) \xleftrightarrow{\mathcal{F}} 1$$

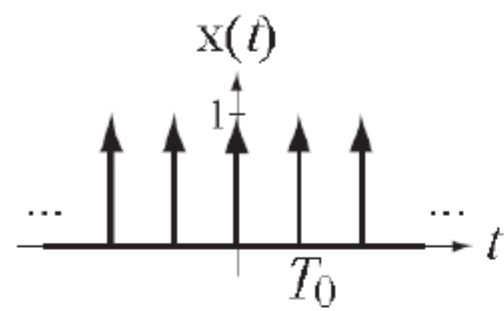


$$1 \xleftrightarrow{\mathcal{F}} \delta(f)$$



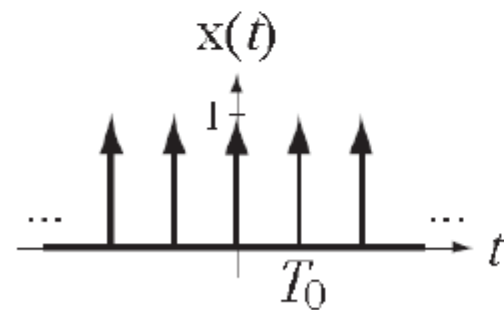
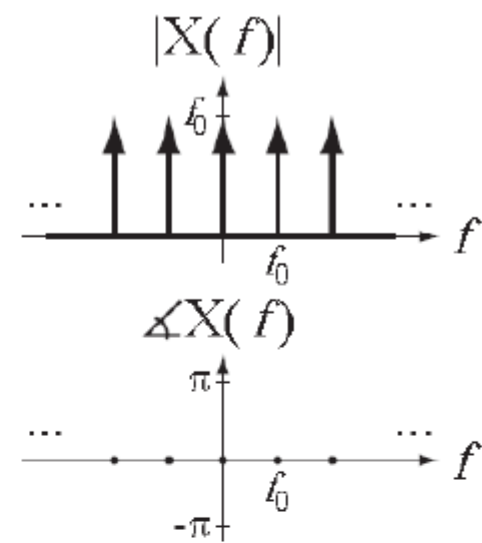
$$1 \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega)$$





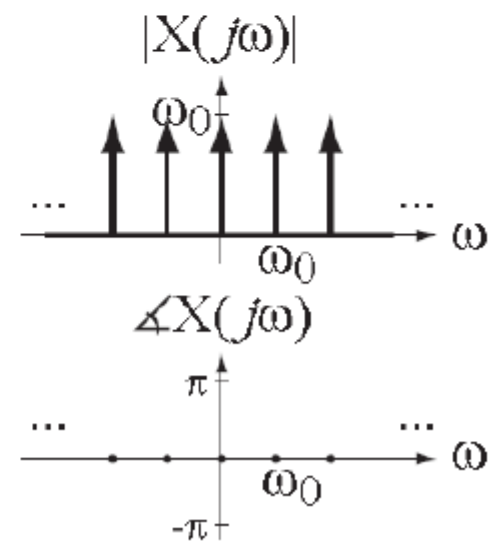
$$\delta_{T_0}(t) \xleftrightarrow{\mathcal{F}} f_0 \delta_{f_0}(f)$$

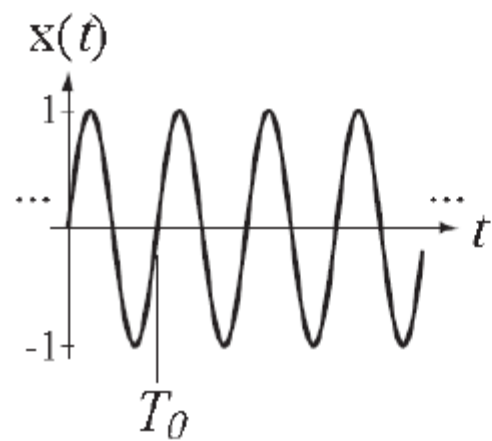
$$f_0 = 1/T_0$$



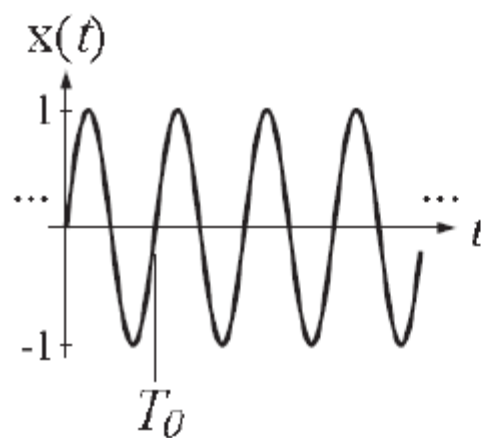
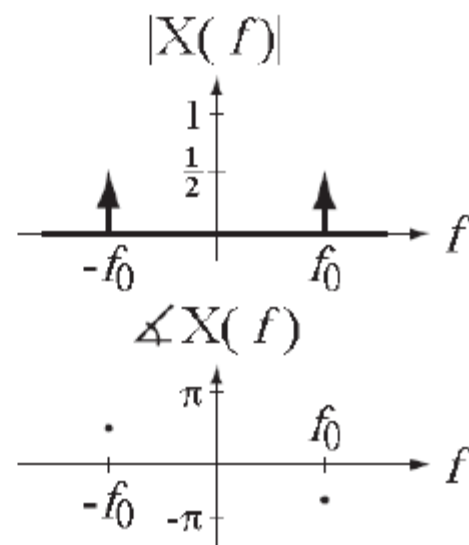
$$\delta_{T_0}(t) \xleftrightarrow{\mathcal{F}} \omega_0 \delta_{\omega_0}(\omega)$$

$$\omega_0 = 2\pi/T_0$$

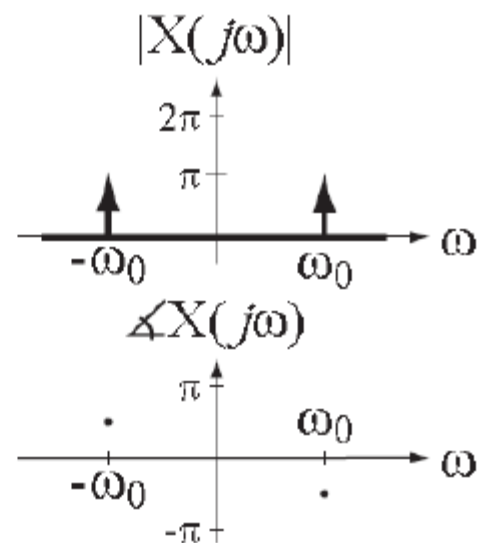


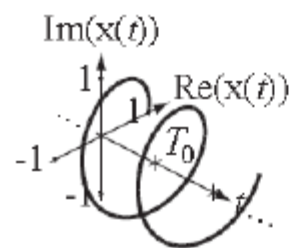


$$\sin(2\pi f_0 t) \xleftrightarrow{\mathcal{F}} \frac{j}{2} [\delta(f + f_0) - \delta(f - f_0)]$$

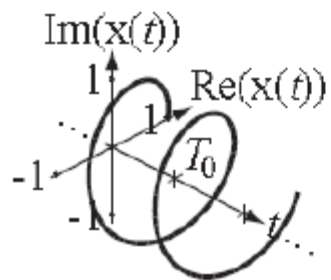
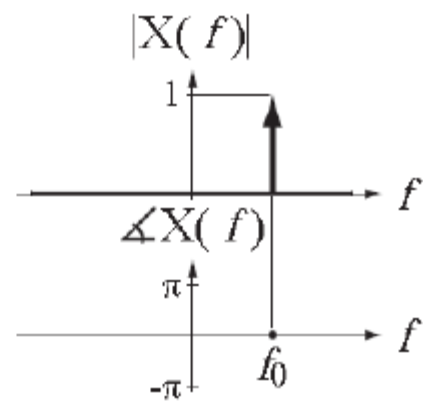


$$\sin(\omega_0 t) \xleftrightarrow{\mathcal{F}} j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

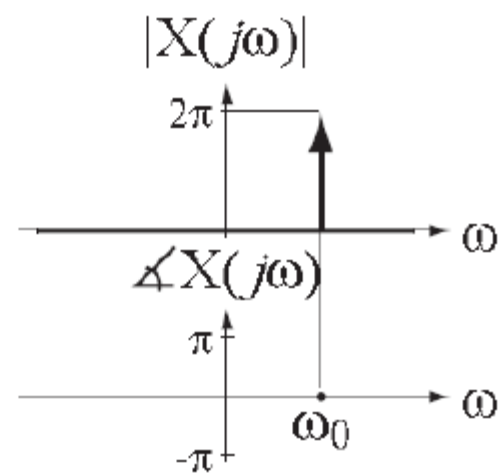


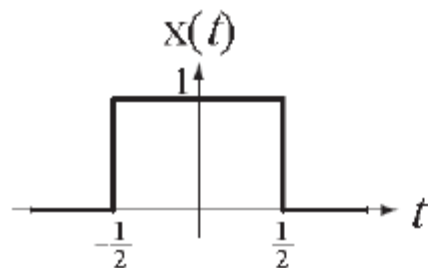


$$e^{j2\pi f_0 t} \xleftrightarrow{\mathcal{F}} \delta(f - f_0)$$



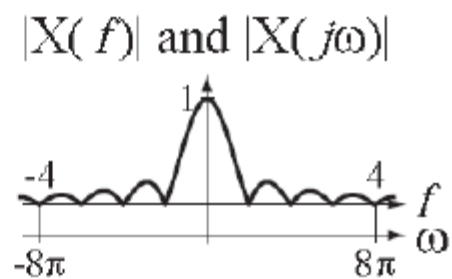
$$e^{j\omega_0 t} \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega - \omega_0)$$



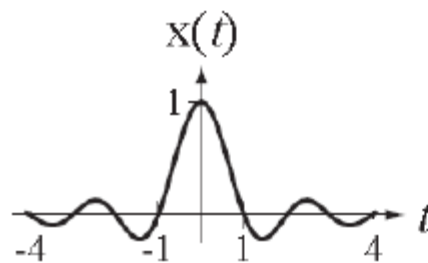
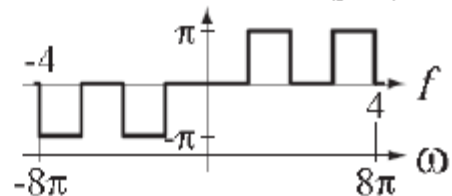


$$\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f)$$

$$\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(\omega/2\pi)$$

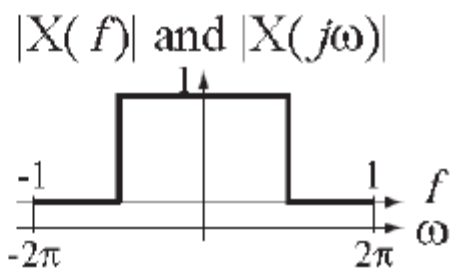


$$\angle X(f) \text{ and } \angle X(j\omega)$$

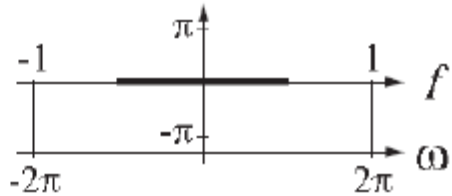


$$\text{sinc}(t) \xleftrightarrow{\mathcal{F}} \text{rect}(f)$$

$$\text{sinc}(t) \xleftrightarrow{\mathcal{F}} \text{rect}(\omega/2\pi)$$

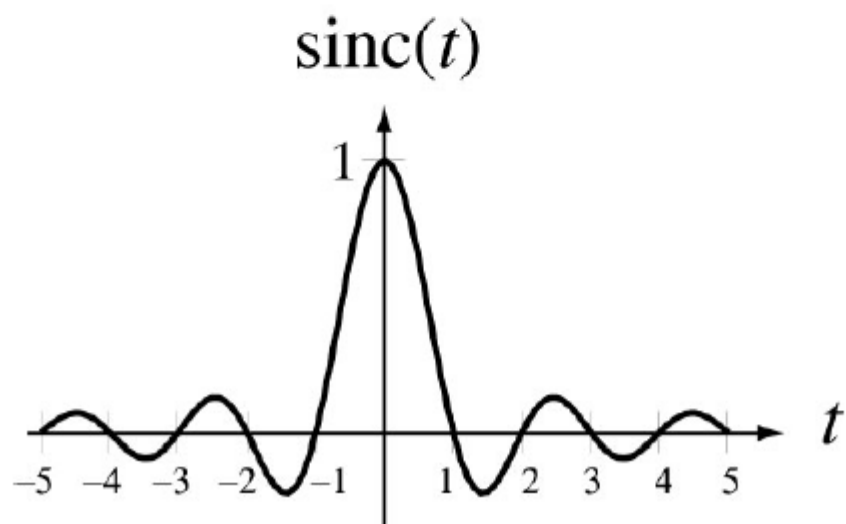


$$\angle X(f) \text{ and } \angle X(j\omega)$$



Sinc Function

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

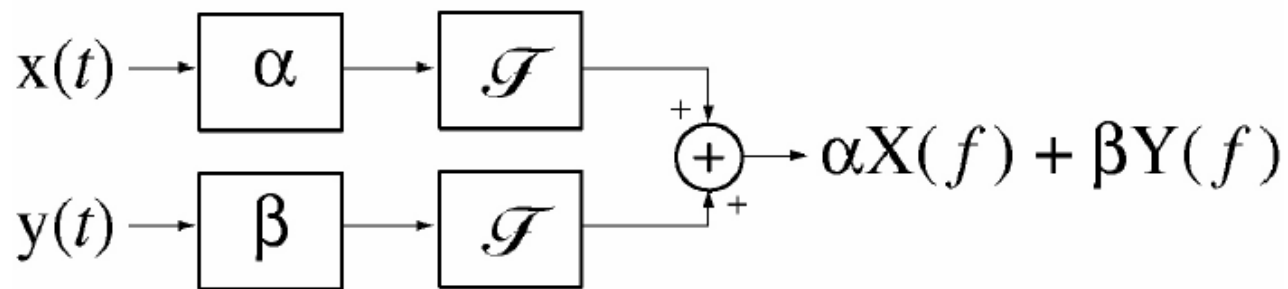
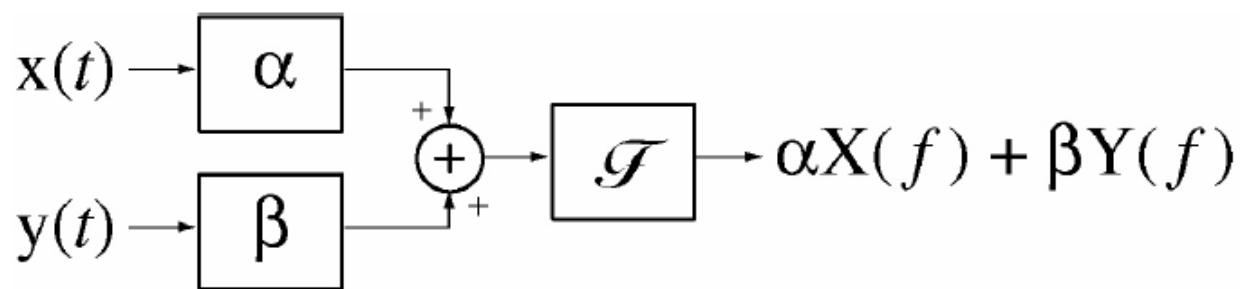


FT Properties

Lineality

$$\alpha x(t) + \beta y(t) \xleftrightarrow{\mathcal{F}} \alpha X(f) + \beta Y(f)$$

$$\alpha x(t) + \beta y(t) \xleftrightarrow{\mathcal{F}} \alpha X(j\omega) + \beta Y(j\omega)$$



Domains Duality

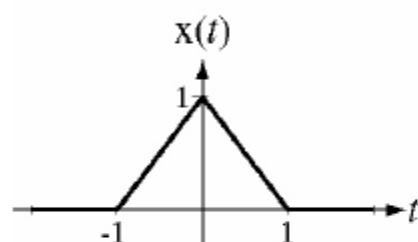
$$X(t) \xleftrightarrow{\mathbb{F}} x(-f) \quad \text{and} \quad X(-t) \xleftrightarrow{\mathbb{F}} x(f)$$

$$X(jt) \xleftrightarrow{\mathbb{F}} 2\pi x(-\omega) \quad \text{and} \quad X(-jt) \xleftrightarrow{\mathbb{F}} 2\pi x(\omega)$$

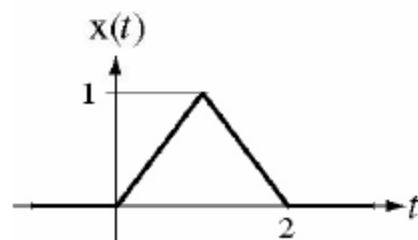
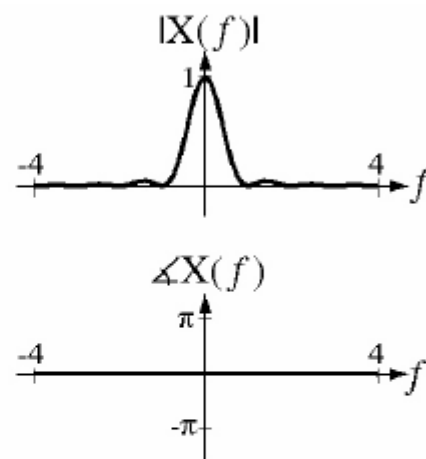
Time Shifting

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} X(f) e^{-j2\pi f t_0}$$

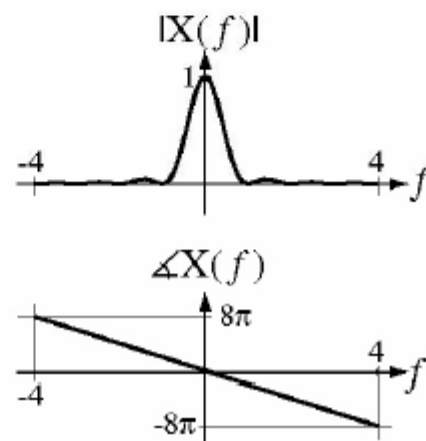
$$x(t - t_0) \xleftrightarrow{\mathcal{F}} X(j\omega) e^{-j\omega t_0}$$



$\xleftrightarrow{\mathcal{F}}$



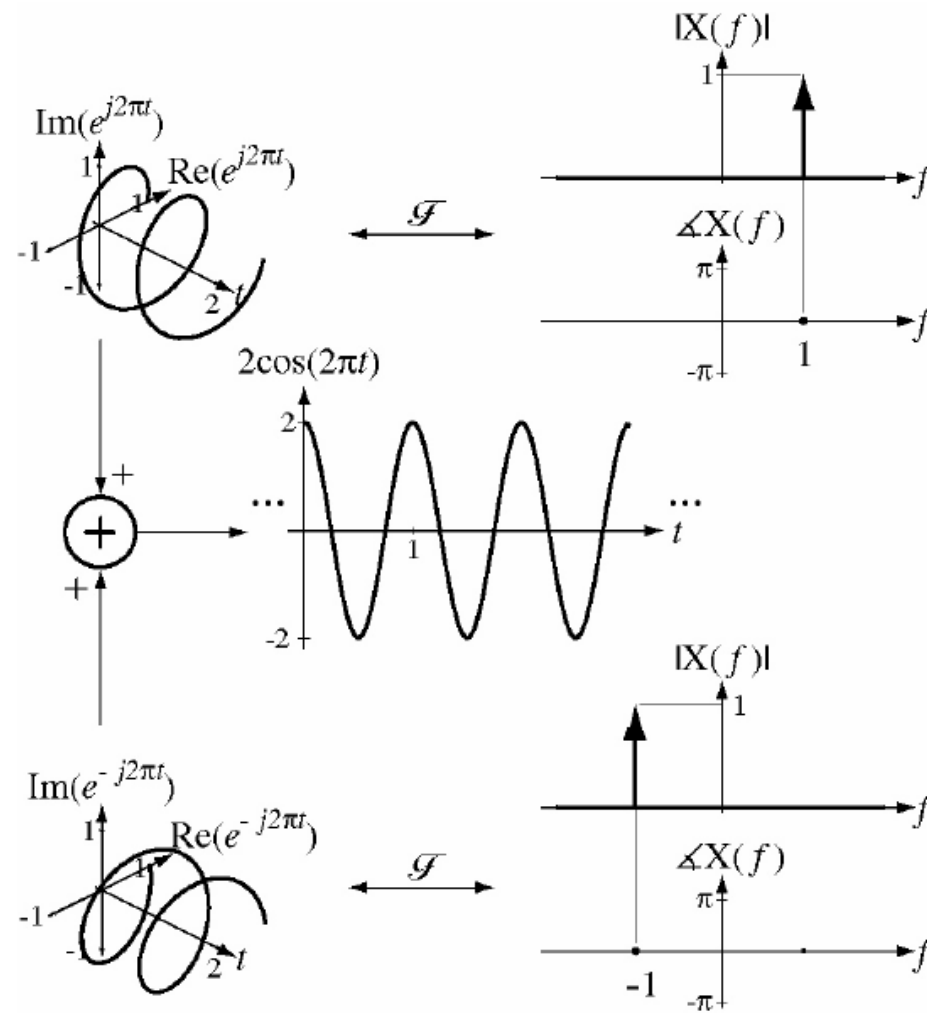
$\xleftrightarrow{\mathcal{F}}$



Frequency Shifting

$$x(t)e^{+j2\pi f_0 t} \xleftrightarrow{\mathcal{F}} X(f - f_0)$$

$$x(t)e^{+j\omega_0 t} \xleftrightarrow{\mathcal{F}} X(j(\omega - \omega_0))$$



Parseval's Theorem

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 df$$

Multiplication Convolution Duality

$$\begin{aligned}x(t) * y(t) &\xleftrightarrow{\mathcal{F}} X(f)Y(f) \\x(t) * y(t) &\xleftrightarrow{\mathcal{F}} X(j\omega)Y(j\omega)\end{aligned}$$

$$\begin{aligned}x(t)y(t) &\xleftrightarrow{\mathcal{F}} X(f) * Y(f) \\x(t)y(t) &\xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} X(j\omega) * Y(j\omega)\end{aligned}$$

Time Domain (t)

Frequency Domain (f)

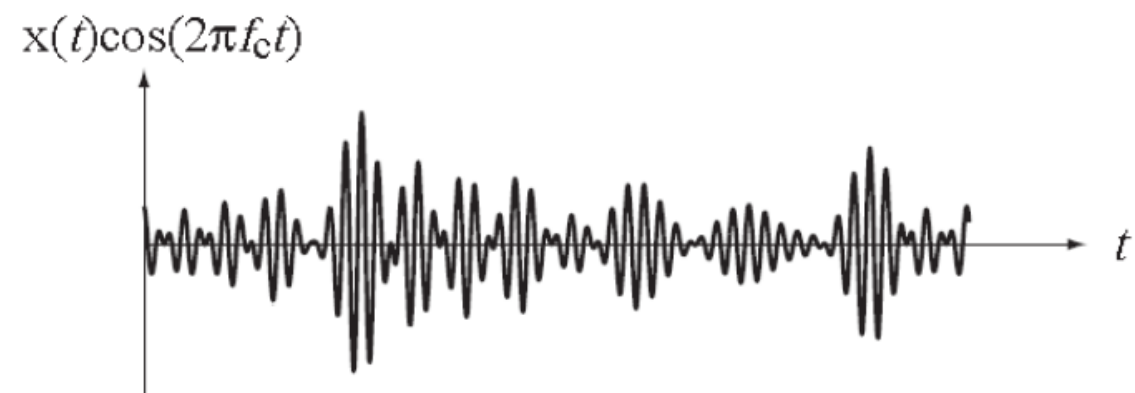
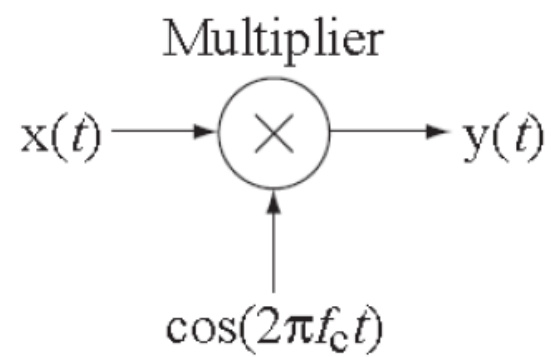
$$y = x * h \longleftrightarrow Y = XH$$

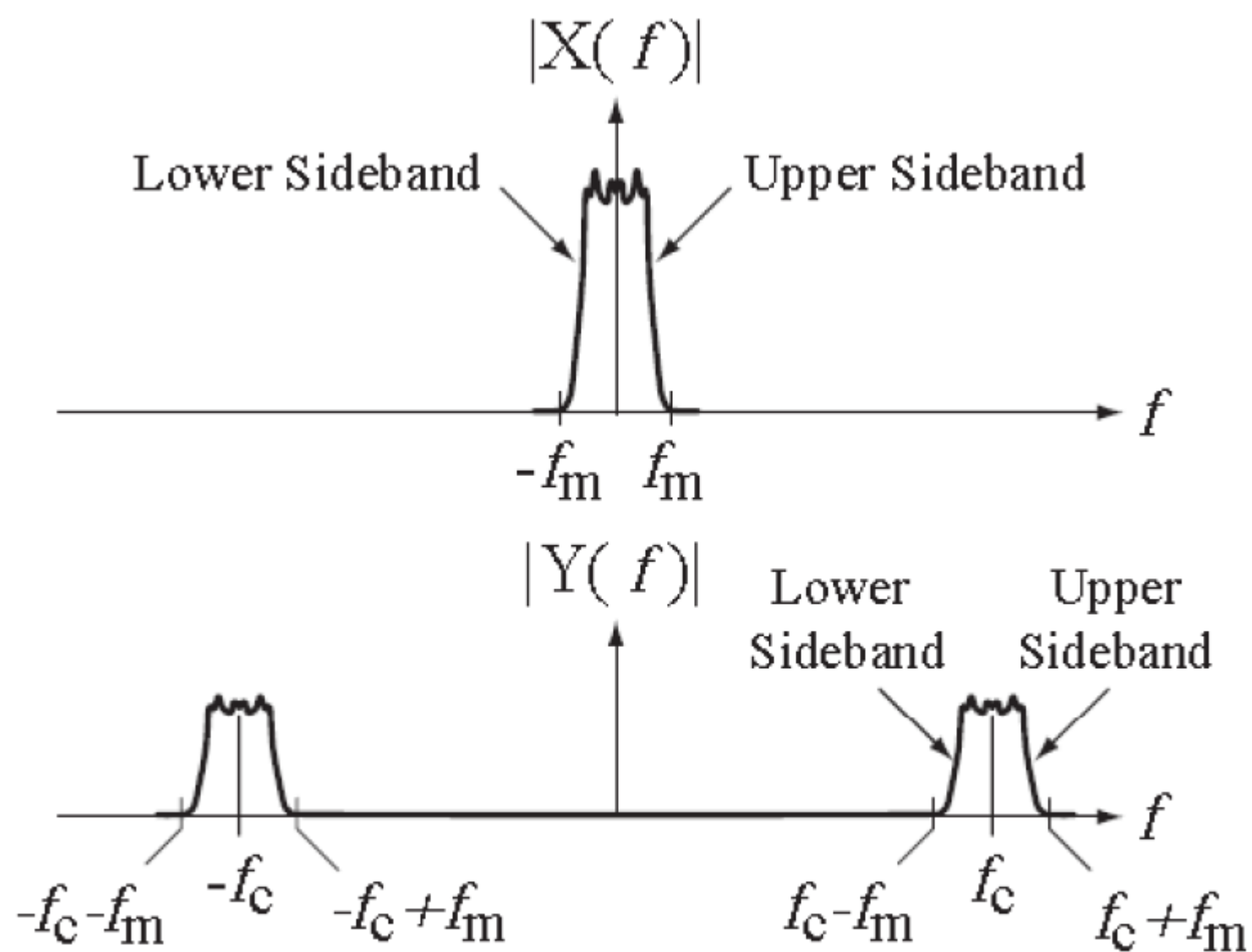
$$y = xh \longleftrightarrow Y = X * H$$

Modulation

$$x(t) \cos(2\pi f_0 t) \xleftrightarrow{\text{F}} \frac{1}{2} [X(f - f_0) + X(f + f_0)]$$

$$x(t) \cos(\omega_0 t) \xleftrightarrow{\text{F}} \frac{1}{2} [X(j(\omega - \omega_0)) + X(j(\omega + \omega_0))]$$





Example

Consider the following system:

$$y(t) = 2x(t) + 3x(t - 2)$$

Answer the following questions.

1. Find the **unit impulse response** $h(t)$ of the system.
2. Find the **frequency response** $H(\omega)$.
3. If the input is

$$x(t) = 2 \cos(2\pi t)$$

find the output $y(t)$.

Step 1 — Find the Impulse Response

For a linear time-invariant (LTI) system, the output is

$$y(t) = x(t) * h(t)$$

The impulse response is the output when the input is a **unit impulse**:

$$x(t) = \delta(t)$$

Substitute into the system equation.

$$y(t) = 2\delta(t) + 3\delta(t - 2)$$

Therefore,

$$h(t) = 2\delta(t) + 3\delta(t - 2)$$

Step 2 — Find the Frequency Response

The frequency response is the Fourier Transform of the impulse response.

$$H(\omega) = \mathcal{F}\{h(t)\}$$

Using the Fourier transform properties:

$$\mathcal{F}\{\delta(t)\} = 1$$

$$\mathcal{F}\{\delta(t - t_0)\} = e^{-j\omega t_0}$$

Apply to each term.

$$H(\omega) = 2(1) + 3e^{-j2\omega}$$

Thus,

$$H(\omega) = 2 + 3e^{-j2\omega}$$

Step 3 — Input Signal

The input is

$$x(t) = 2 \cos(2\pi t)$$

Angular frequency is

$$\omega_0 = 2\pi$$

For LTI systems, a cosine input produces a cosine output:

$$y(t) = A|H(\omega_0)| \cos(\omega_0 t + \angle H(\omega_0))$$

where

- A = input amplitude
- $H(\omega_0)$ = frequency response at that frequency.

Step 4 — Evaluate the Frequency Response

Compute

$$H(2\pi) = 2 + 3e^{-j2(2\pi)}$$

$$H(2\pi) = 2 + 3e^{-j4\pi}$$

Since

$$e^{-j4\pi} = 1$$

we get

$$H(2\pi) = 2 + 3 = 5$$

4. Integrals

Formula for integration by parts: $\int_a^b u \frac{dv}{dx} dx = [uv]_a^b - \int_a^b \frac{du}{dx} v dx$

$f(x)$	$\int f(x) dx$	$f(x)$	$\int f(x) dx$
x^n	$\frac{x^{n+1}}{n+1} \quad (n \neq -1)$	$[g(x)]^n g'(x)$	$\frac{[g(x)]^{n+1}}{n+1} \quad (n \neq -1)$
$\frac{1}{x}$	$\ln x $	$\frac{g'(x)}{g(x)}$	$\ln g(x) $
e^x	e^x	a^x	$\frac{a^x}{\ln a} \quad (a > 0)$
$\sin x$	$-\cos x$	$\sinh x$	$\cosh x$
$\cos x$	$\sin x$	$\cosh x$	$\sinh x$
$\tan x$	$-\ln \cos x $	$\tanh x$	$\ln \cosh x$
$\operatorname{cosec} x$	$\ln \left \tan \frac{x}{2} \right $	$\operatorname{cosech} x$	$\ln \left \tanh \frac{x}{2} \right $
$\sec x$	$\ln \sec x + \tan x $	$\operatorname{sech} x$	$2 \tan^{-1} e^x$
$\sec^2 x$	$\tan x$	$\operatorname{sech}^2 x$	$\tanh x$
$\cot x$	$\ln \sin x $	$\coth x$	$\ln \sinh x $
$\sin^2 x$	$\frac{x}{2} - \frac{\sin 2x}{4}$	$\sinh^2 x$	$\frac{\sinh 2x}{4} - \frac{x}{2}$
$\cos^2 x$	$\frac{x}{2} + \frac{\sin 2x}{4}$	$\cosh^2 x$	$\frac{\sinh 2x}{4} + \frac{x}{2}$

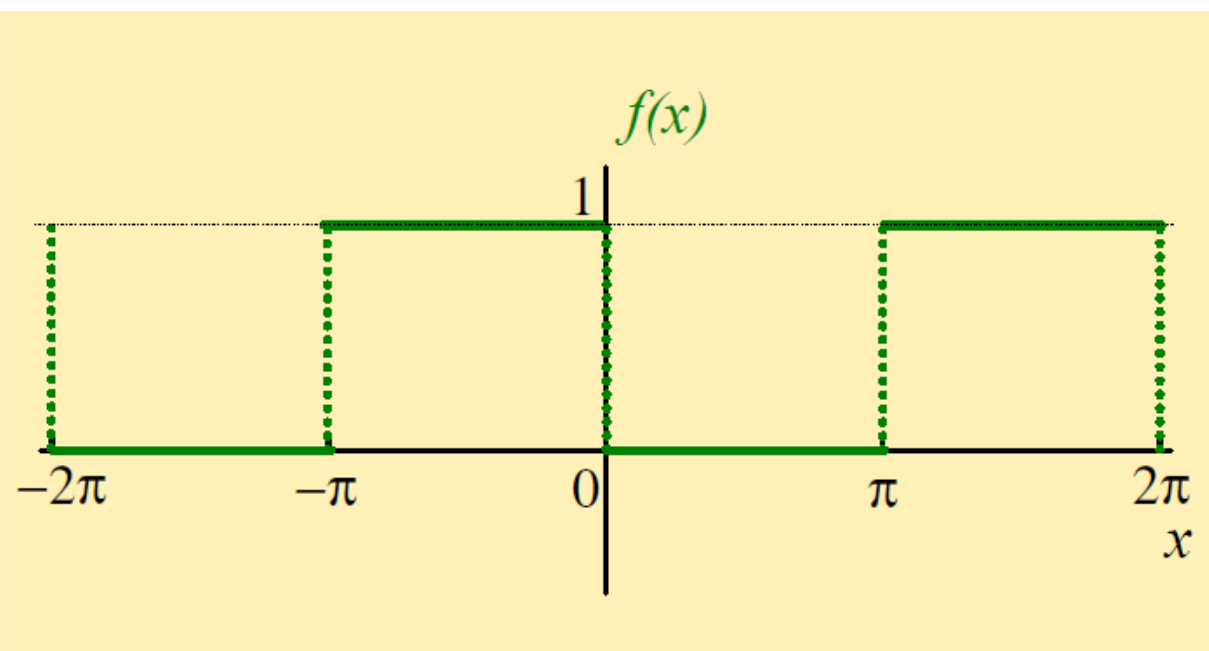
$f(x)$	$\int f(x) dx$	$f(x)$	$\int f(x) dx$
$\frac{1}{a^2+x^2}$	$\frac{1}{a} \tan^{-1} \frac{x}{a}$ $(a > 0)$	$\frac{1}{a^2-x^2}$ $\frac{1}{x^2-a^2}$	$\frac{1}{2a} \ln \left \frac{a+x}{a-x} \right \quad (0 < x < a)$ $\frac{1}{2a} \ln \left \frac{x-a}{x+a} \right \quad (x > a > 0)$
$\frac{1}{\sqrt{a^2-x^2}}$	$\sin^{-1} \frac{x}{a}$ $(-a < x < a)$	$\frac{1}{\sqrt{a^2+x^2}}$ $\frac{1}{\sqrt{x^2-a^2}}$	$\ln \left \frac{x+\sqrt{a^2+x^2}}{a} \right \quad (a > 0)$ $\ln \left \frac{x+\sqrt{x^2-a^2}}{a} \right \quad (x > a > 0)$
$\sqrt{a^2-x^2}$	$\frac{a^2}{2} \left[\sin^{-1} \left(\frac{x}{a} \right) + \frac{x\sqrt{a^2-x^2}}{a^2} \right]$	$\sqrt{a^2+x^2}$ $\sqrt{x^2-a^2}$	$\frac{a^2}{2} \left[\sinh^{-1} \left(\frac{x}{a} \right) + \frac{x\sqrt{a^2+x^2}}{a^2} \right]$ $\frac{a^2}{2} \left[-\cosh^{-1} \left(\frac{x}{a} \right) + \frac{x\sqrt{x^2-a^2}}{a^2} \right]$

Full worked solutions

Exercise 1.

$$f(x) = \begin{cases} 1, & -\pi < x < 0 \\ 0, & 0 < x < \pi, \end{cases} \quad \text{and has period } 2\pi$$

a) Sketch a graph of $f(x)$ in the interval $-2\pi < x < 2\pi$



b) Fourier series representation of $f(x)$

STEP ONE

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 f(x) dx + \frac{1}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 1 \cdot dx + \frac{1}{\pi} \int_0^{\pi} 0 \cdot dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 dx$$

$$= \frac{1}{\pi} [x]_{-\pi}^0$$

$$= \frac{1}{\pi} (0 - (-\pi))$$

$$= \frac{1}{\pi} \cdot (\pi)$$

$$\text{i.e. } a_0 = 1 .$$

STEP TWO

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{\pi} \int_{-\pi}^0 f(x) \cos nx \, dx + \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx \, dx \\&= \frac{1}{\pi} \int_{-\pi}^0 1 \cdot \cos nx \, dx + \frac{1}{\pi} \int_0^{\pi} 0 \cdot \cos nx \, dx \\&= \frac{1}{\pi} \int_{-\pi}^0 \cos nx \, dx \\&= \frac{1}{\pi} \left[\frac{\sin nx}{n} \right]_{-\pi}^0 = \frac{1}{n\pi} [\sin nx]_{-\pi}^0 \\&= \frac{1}{n\pi} (\sin 0 - \sin(-n\pi)) \\&= \frac{1}{n\pi} (0 + \sin n\pi) \\ \text{i.e. } a_n &= \frac{1}{n\pi} (0 + 0) = 0.\end{aligned}$$

STEP THREE

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \\
 &= \frac{1}{\pi} \int_{-\pi}^0 f(x) \sin nx \, dx + \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx \, dx \\
 &= \frac{1}{\pi} \int_{-\pi}^0 1 \cdot \sin nx \, dx + \frac{1}{\pi} \int_0^{\pi} 0 \cdot \sin nx \, dx
 \end{aligned}$$

$$\begin{aligned}
 \text{i.e. } b_n &= \frac{1}{\pi} \int_{-\pi}^0 \sin nx \, dx = \frac{1}{\pi} \left[\frac{-\cos nx}{n} \right]_{-\pi}^0 \\
 &= -\frac{1}{n\pi} [\cos nx]_{-\pi}^0 = -\frac{1}{n\pi} (\cos 0 - \cos(-n\pi)) \\
 &= -\frac{1}{n\pi} (1 - \cos n\pi) = -\frac{1}{n\pi} (1 - (-1)^n), \text{ see } \text{TRIG}
 \end{aligned}$$

$$\text{i.e. } b_n = \begin{cases} 0 & , n \text{ even} \\ -\frac{2}{n\pi} & , n \text{ odd} \end{cases}, \text{ since } (-1)^n = \begin{cases} 1 & , n \text{ even} \\ -1 & , n \text{ odd} \end{cases}$$

Solutions to exercises

We now have that

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

with the three steps giving

$$a_0 = 1, \quad a_n = 0, \quad \text{and} \quad b_n = \begin{cases} 0 & , n \text{ even} \\ -\frac{2}{n\pi} & , n \text{ odd} \end{cases}$$

It may be helpful to construct a table of values of b_n

n	1	2	3	4	5
b_n	$-\frac{2}{\pi}$	0	$-\frac{2}{\pi} \left(\frac{1}{3}\right)$	0	$-\frac{2}{\pi} \left(\frac{1}{5}\right)$

Substituting our results now gives the required series

$$f(x) = \frac{1}{2} - \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right]$$

c) Pick an appropriate value of x , to show that

$$\boxed{\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots}$$

Comparing this series with

$$f(x) = \frac{1}{2} - \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right],$$

we need to introduce a minus sign in front of the constants $\frac{1}{3}, \frac{1}{7}, \dots$

So we need $\sin x = 1, \sin 3x = -1, \sin 5x = 1, \sin 7x = -1$, etc

The first condition of $\sin x = 1$ suggests trying $x = \frac{\pi}{2}$.

$$\begin{array}{ccccccc} \text{This choice gives} & \sin \frac{\pi}{2} & + & \frac{1}{3} \sin 3\frac{\pi}{2} & + & \frac{1}{5} \sin 5\frac{\pi}{2} & + & \frac{1}{7} \sin 7\frac{\pi}{2} \\ \text{i.e.} & 1 & - & \frac{1}{3} & + & \frac{1}{5} & - & \frac{1}{7} \end{array}$$

Looking at the graph of $f(x)$, we also have that $f(\frac{\pi}{2}) = 0$.

Picking $x = \frac{\pi}{2}$ thus gives

$$0 = \frac{1}{2} - \frac{2}{\pi} \left[\sin \frac{\pi}{2} + \frac{1}{3} \sin \frac{3\pi}{2} + \frac{1}{5} \sin \frac{5\pi}{2} + \frac{1}{7} \sin \frac{7\pi}{2} + \dots \right]$$

$$\text{i.e. } 0 = \frac{1}{2} - \frac{2}{\pi} \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right]$$

A little manipulation then gives a series representation of $\frac{\pi}{4}$

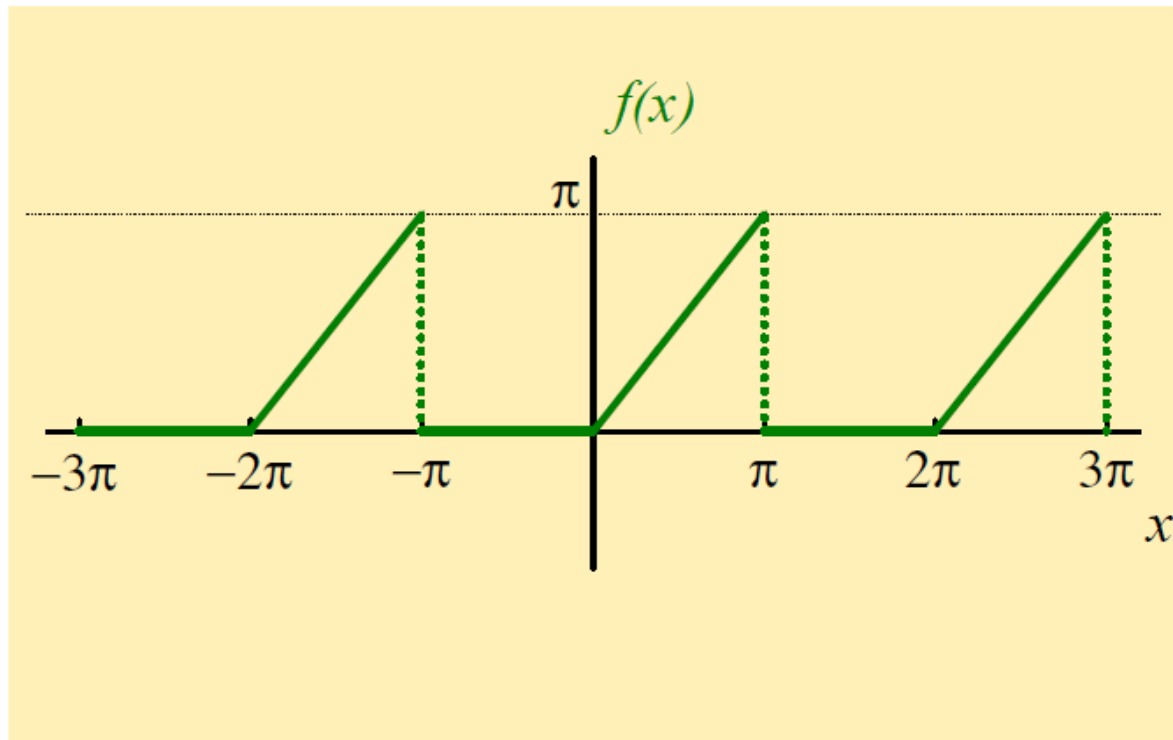
$$\frac{2}{\pi} \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right] = \frac{1}{2}$$
$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}.$$

[Return to Exercise 1](#)

Exercise 2.

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi, \end{cases} \quad \text{and has period } 2\pi$$

a) Sketch a graph of $f(x)$ in the interval $-3\pi < x < 3\pi$



b) Fourier series representation of $f(x)$

STEP ONE

$$\begin{aligned}a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \mathrm{d}x &= \frac{1}{\pi} \int_{-\pi}^0 f(x) \mathrm{d}x + \frac{1}{\pi} \int_0^{\pi} f(x) \mathrm{d}x \\&= \frac{1}{\pi} \int_{-\pi}^0 0 \cdot \mathrm{d}x + \frac{1}{\pi} \int_0^{\pi} x \mathrm{d}x \\&= \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} \\&= \frac{1}{\pi} \left(\frac{\pi^2}{2} - 0 \right) \\&\text{i.e. } a_0 = \frac{\pi}{2} .\end{aligned}$$

Homework

Problem 1. Determine which of the following signals are periodic. If a signal is periodic, determine its fundamental period.

(a) $y(t) = \sin(\frac{5\pi}{4}t)$

(c) $y(t) = \cos(\frac{3}{5}t) + |\sin(\frac{5}{3}t)|$

(b) $y(t) = e^{j(t+\frac{\pi}{3})}$

Problem 13. Determine which of the following signals are periodic. If a signal is periodic, determine its fundamental period.

(a) $x[n] = \sin(\frac{7\pi}{3}n)$

(c) $x[n] = e^{j(n+3)}$

(b) $x[n] = \cos(n\pi) + \cos(n)$

(d) $x[n] = \cos(\frac{\pi}{4}n^2)$

Problem 14. Sketch the following signal and determine its fundamental period.

Problem 6. Evaluate the following expressions using the unit impulse and unit step functions properties:

(a) $\int_0^3 \sin(t) \delta(3t - 6) dt$

(d) $\int_{-\infty}^{\infty} \delta(4 - t^2) dt$

(b) $\int_0^4 2 \cos(t - 2) \delta'(t - 2) dt$

(c) $x(t) = t \delta'(t) + \delta(t)$

(e) $x(t) = \int_{t-1}^{t-4} \tau^2 \delta(t - \tau - 2) d\tau$

Problem 8. A continuous-time signal $x(t)$ is shown in Fig. 1.8. Sketch and label each of the following signals.

(a) $y(t) = x(t) u(t) + x(-t) u(t)$

(b) $y(t) = x(2t - 1)$

(c) $y(t) = x(-\frac{t}{2} + 1)$

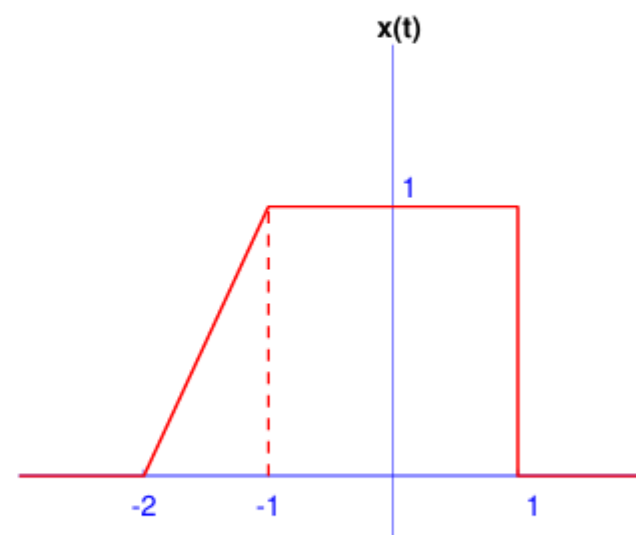
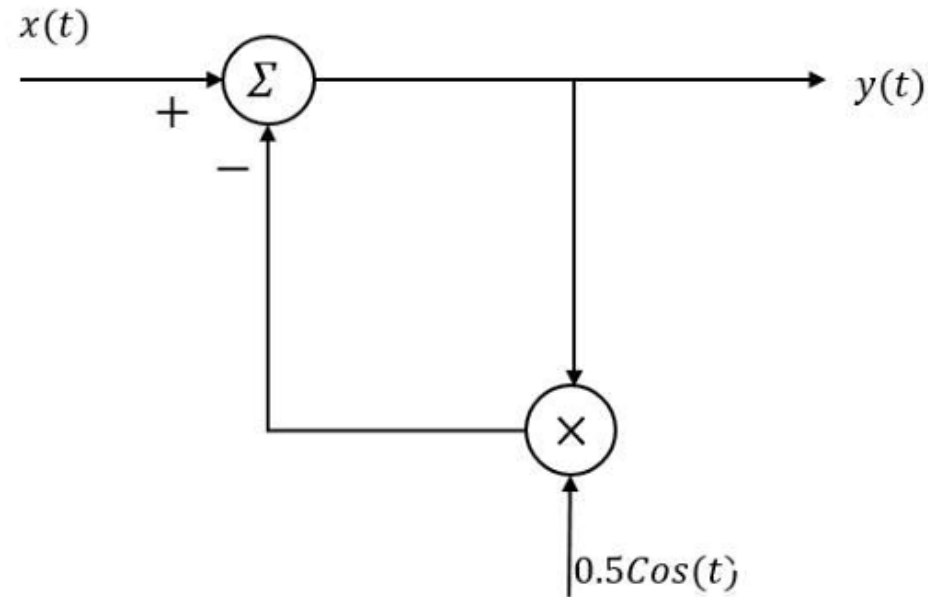


Figure 1.8: Problem 8.

Problem 28. Consider the continuous-time system shown in Fig. 2.1.



- (b) Determine whether the system is (i) LTI, (ii) causal, (ii) memoryless, (iv) invertible, (v) stable.
- (a) Find the relationship between the input and the output.

Problem 47. Consider the system initially at rest condition, shown in Fig. 3.11.

- (a) Find the input-output relationship for this system.
- (b) Find the impulse response $h(t)$ of this system.
- (c) Is this system memoryless?
- (d) Is this system causal?

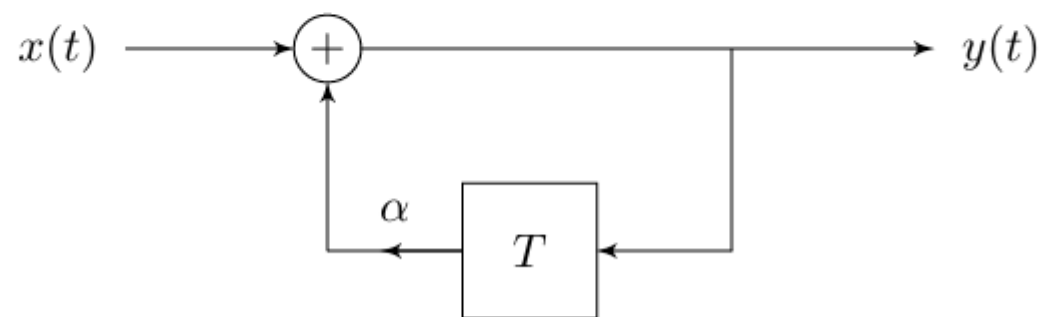


Figure 3.5: Problem 47.

Problem 50. A system is formed by connecting 3 systems as shown in Fig. 3.12. The impulse responses of the systems are given by

$$h_1[n] = \delta[n] + 3\delta[n - 1], \quad h_2[n] = u[n] \quad \text{and} \quad h_3[n] = \delta[n - 3].$$

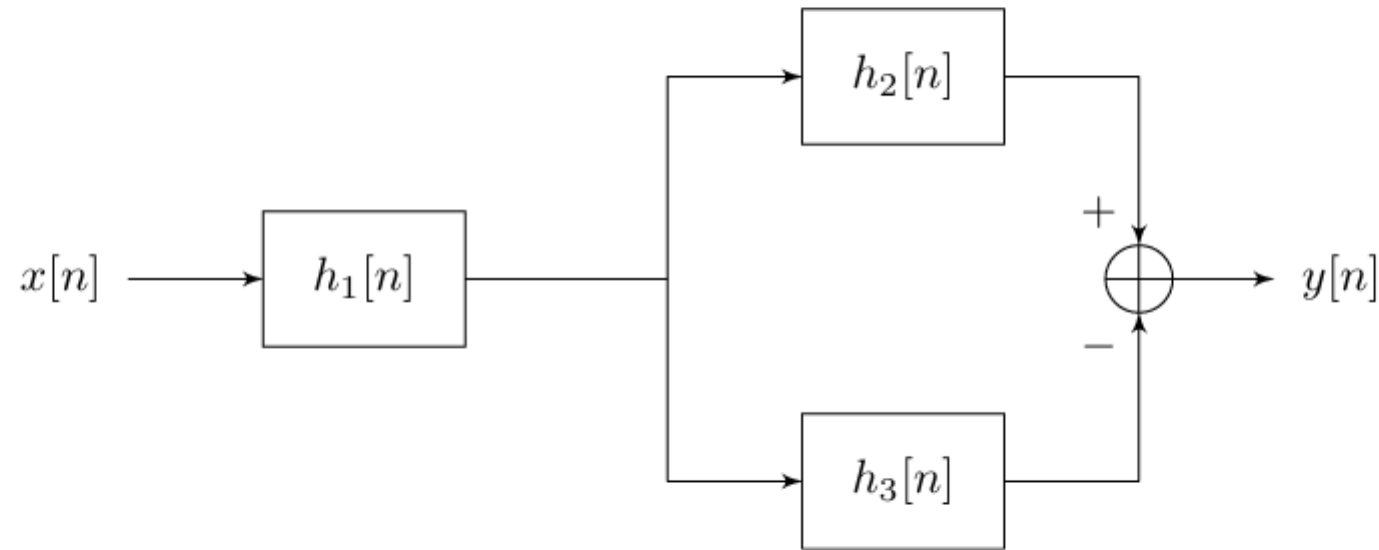


Figure 3.6: Problem 50.

Find the impulse response $h[n]$ of the overall system.